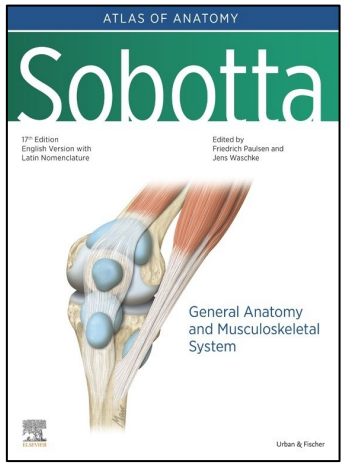
Modul 5.3 Bevægeapparatet

Columnas (COL) - regionale anatomi





Tong Tong
Assistant Professor

Normal text = Danish
Normal text + Italic = Latin anatomical term
Italic + smaller + green = English

Part I — Spinalkanalen og spinalnerver

Part I – Vertebral Canal and Spinal Nerves

Læringsmål

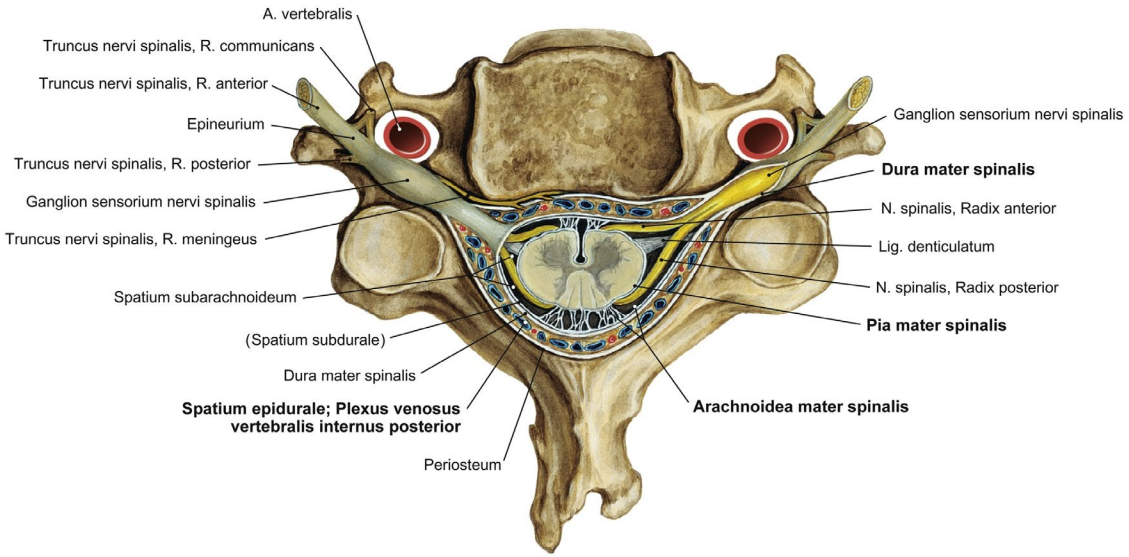
- Redegøre for spinalkanalens indhold, rygmarvshinderne og rygmarvens afslutning.
  Describe the contents of the vertebral canal, spinal meninges, and termination of the spinal cord.

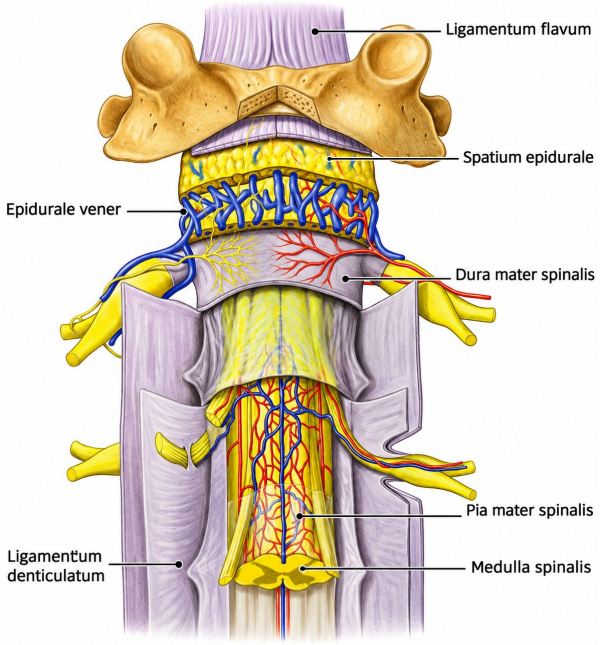
- Forklare spinalnervernes rødder, forgrening og forløb i relation til rygsøjlen og foramina intervertebralia.
  Explain the roots, branches, and course of spinal nerves in relation to the vertebral column and intervertebral foramina.

- Anvende den regionale anatomi til at forklare nerverodskompression og det anatomiske grundlag for lumbalpunktur.
  Apply regional anatomy to explain nerve-root compression and the anatomical basis of lumbar puncture.

Spinalkanalen – Indhold og rygmarvshinder

Vertebral Canal – Contents and Spinal Meninges





- Spinalkanalen (***canalis vertebralis***): Indeholder rygmarven, spinalnerverødder, hinder og blodkar.
Vertebral canal: Contains the spinal cord, spinal nerve roots, meninges, and blood vessels.

- Rygmarvshinder: Dura mater, arachnoidea mater og pia mater omgiver rygmarven.
Spinal meninges: Dura mater, arachnoid mater, and pia mater surround the spinal cord.

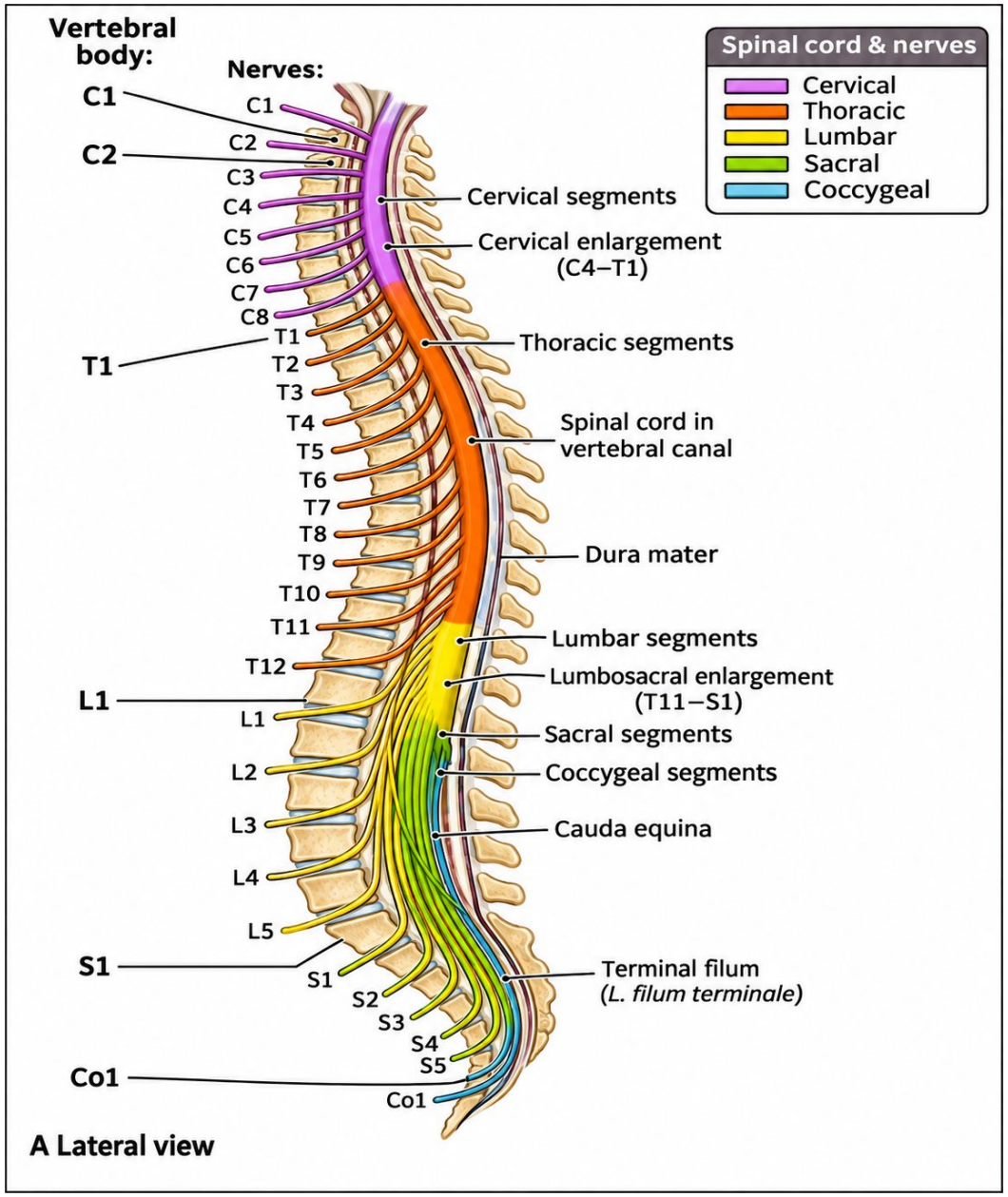
- Rum omkring rygmarven: Epiduralrummet indeholder fedt og venøse plexus; subaraknoidalrummet indeholder cerebrospinalvæske.
Spaces around the spinal cord: The epidural space contains fat and venous plexuses; the subarachnoid space contains cerebrospinal fluid.

Rygmarvens segmenter og spinalnervernes forløb

Spinal Cord Segments and Course of Spinal Nerves

- Segmenter og hvirvler: Rygmarvens segmenter svarer ikke altid til niveauet for de tilsvarende hvirvler.
  Spinal cord segments do not always correspond to the levels of the identically numbered vertebrae.

- Nerverødder: Løber gradvist længere nedad, før de forlader rygsøjlen.
  Nerve roots descend progressively farther before exiting the vertebral column.



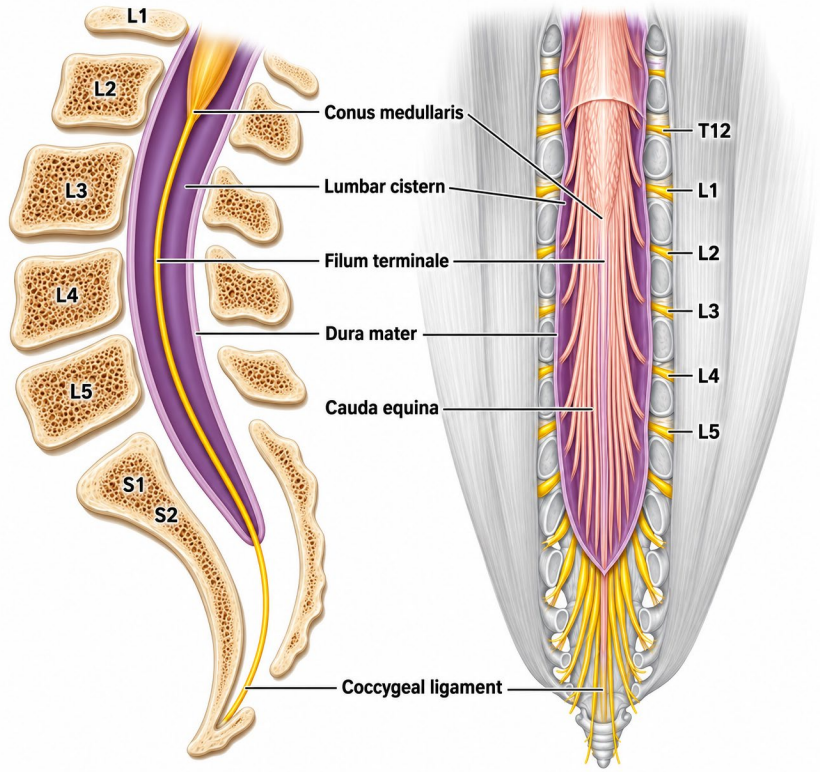
Rygmarvens afslutning og cauda equina

Spinal Cord Termination and Cauda Equina

- Rygmarvskeglen (***conus medullaris***): Rygmarven ender typisk ved L1–L2 hos voksne.
Conus medullaris: The spinal cord typically terminates at L1–L2 in adults.

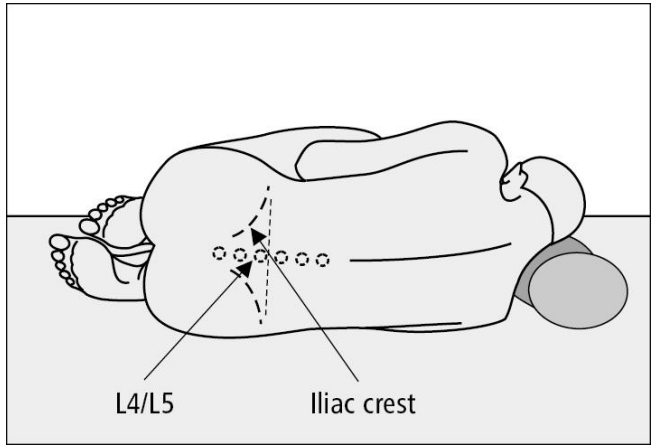
- Hestehalen (***cauda equina***): Lumbale, sakrale og coccygeale nerverødder fortsætter kaudalt i spinalkanalen.
Cauda equina: Lumbar, sacral, and coccygeal nerve roots descend within the vertebral canal.

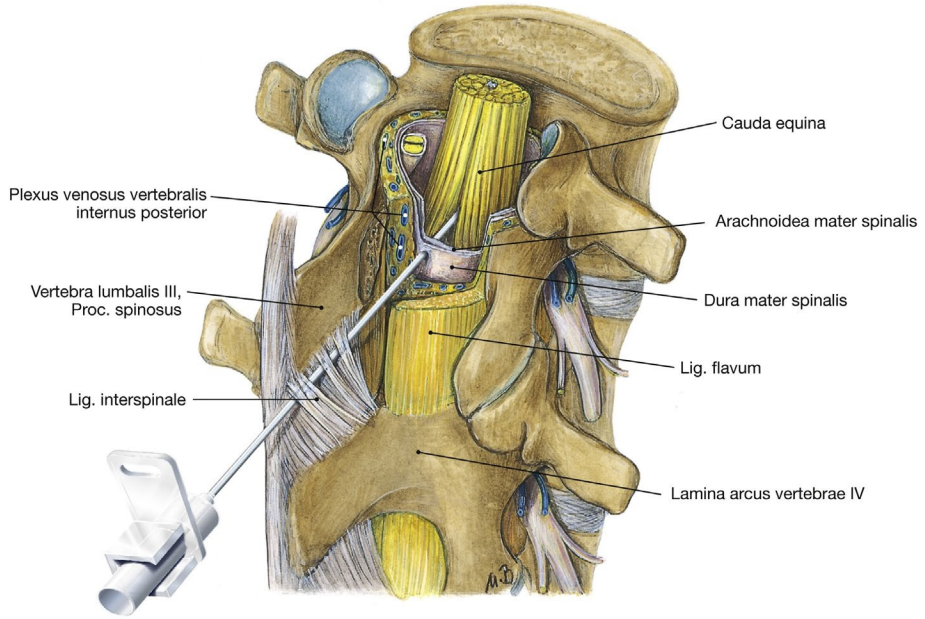
- Durasæk og endetråd: Durasækken ender typisk ved S2; filum terminale bidrager til kaudal forankring.
Dural sac and filum terminale: The dural sac typically ends at S2; the filum terminale contributes to caudal anchoring.

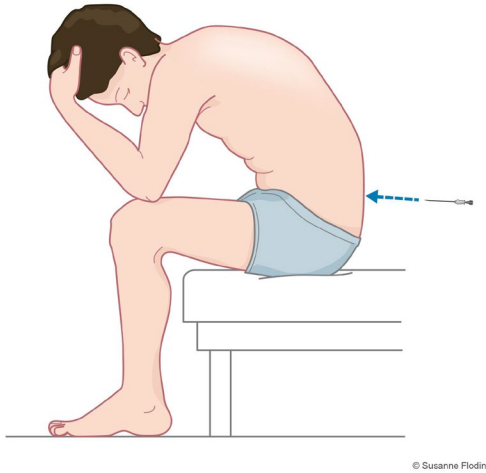


Lumbalpunktur – Anatomisk grundlag

Lumbar Puncture – Anatomical Principles







- Indstiksniveau: Lumbalpunktur udføres typisk mellem L3–L4 eller L4–L5, under rygmarvens afslutning.
Insertion level: Lumbar puncture is typically performed at L3–L4 or L4–L5, below the termination of the spinal cord.

- Anatomisk passage: Nålen passerer gennem de posteriore væv og rygmarvshinderne til spatium subarachnoideum.
Anatomical pathway: The needle passes through posterior tissues and meninges into the subarachnoid space.

- Formål: Cerebrospinalvæske udtages fra spatium subarachnoideum, hvor cauda equina findes.
Purpose: Cerebrospinal fluid is collected from the subarachnoid space containing the cauda equina.

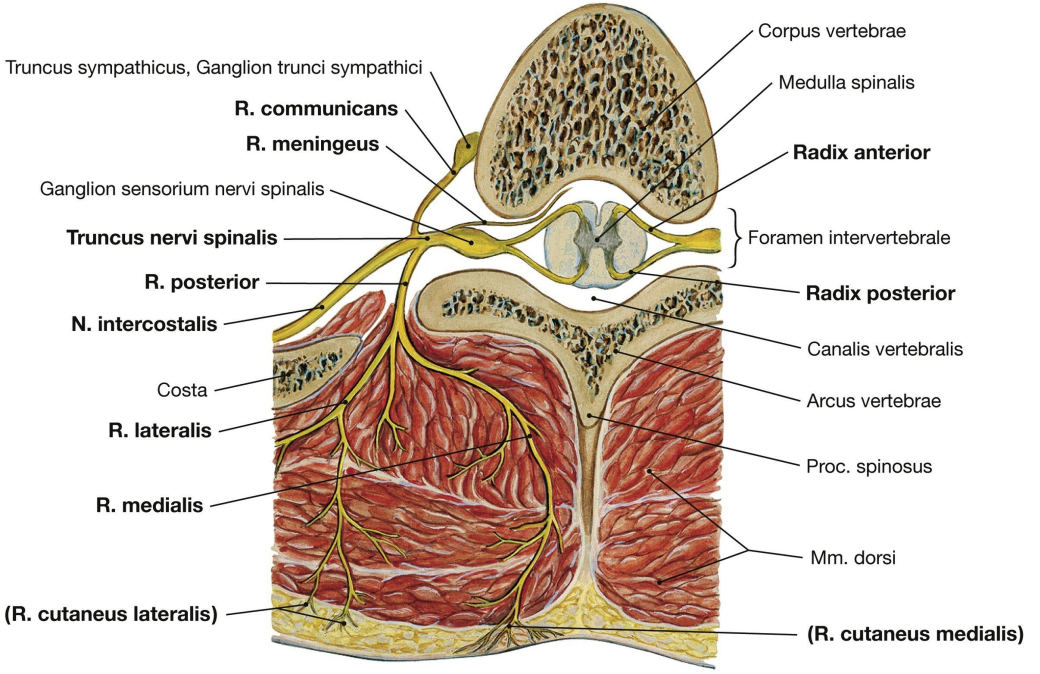
Nerverødder og spinalganglion

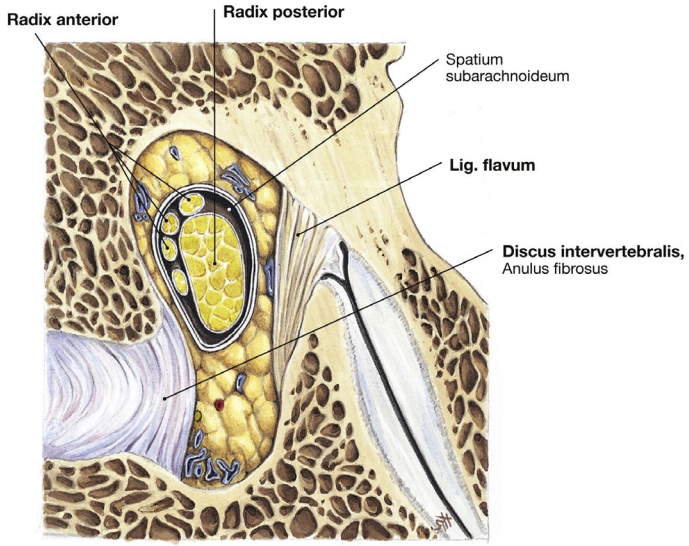
Spinal Nerve Roots and Dorsal Root Ganglion

- Bageste rod (***radix posterior***): Indeholder sensoriske nervefibre, der leder information til rygmarven.
Posterior root: Contains sensory nerve fibers carrying information toward the spinal cord.

- Forreste rod (***radix anterior***): Indeholder motoriske nervefibre, der leder signaler fra rygmarven.
Anterior root: Contains motor nerve fibers carrying signals away from the spinal cord.

- Spinalganglion (***ganglion spinale***): Indeholder sensoriske nervecellelegemer og ligger typisk i eller nær foramen intervertebrale.
Dorsal root ganglion: Contains sensory neuron cell bodies and is usually located within or near the intervertebral foramen.





Spinalnerven og dens forgrening

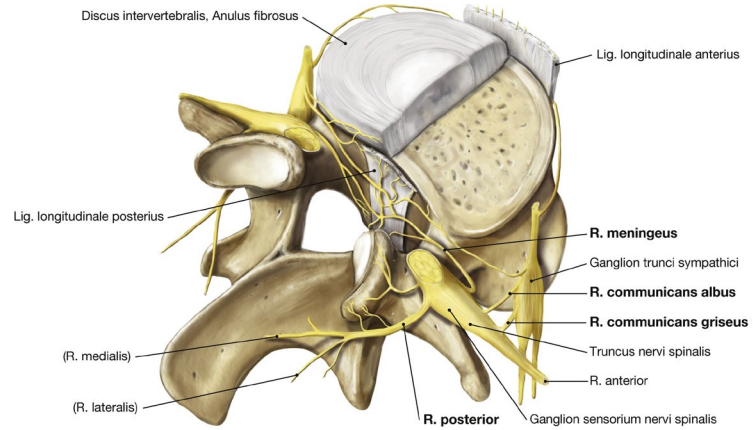
Spinal Nerve and Its Branches

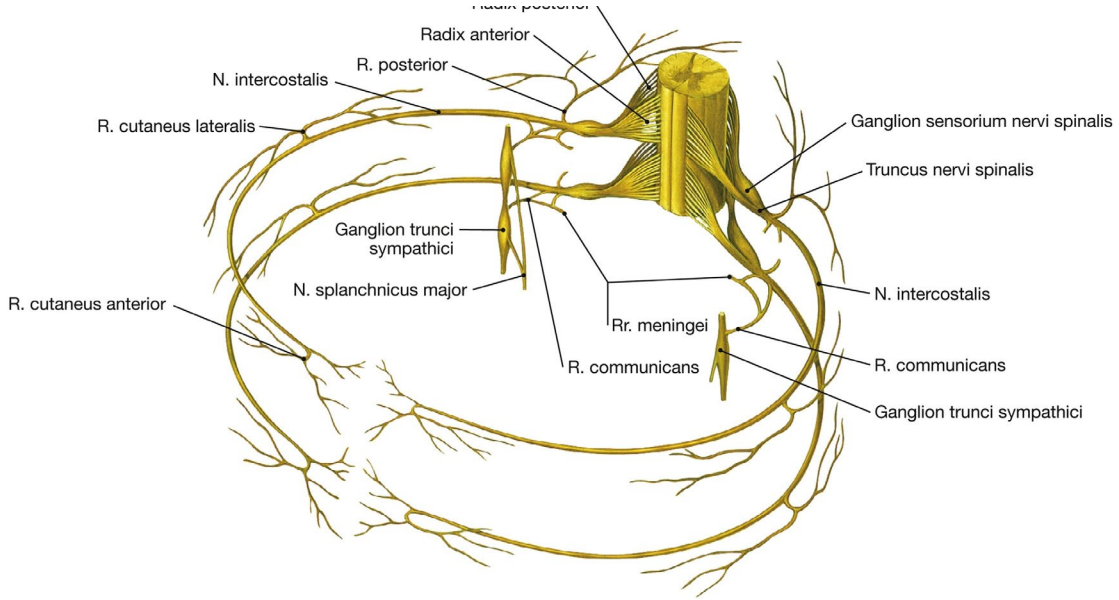
- Spinalnerven (***n. spinalis***): Dannes ved sammenløb af de forreste og bageste nerverødder og er en blandet nerve.
Spinal nerve: Formed by the union of anterior and posterior roots; contains both sensory and motor fibers.

- Bageste gren (***ramus posterior***): Innerverer de dybe rygmuskler, facetled og dele af ryggens hud.
Posterior ramus: Supplies intrinsic back muscles, facet joints, and parts of the skin of the back.

- Forreste gren (***ramus anterior***): Forsyner den anterolaterale kropsvæg og ekstremiteterne; danner nerveplexus i flere regioner.
Anterior ramus: Supplies the anterolateral body wall and limbs; forms nerve plexuses in several regions.

- Forbindelsesgrene (***rami communicantes***): Forbinder spinalnerven med den sympatiske grænsestreng (truncus sympathicus).
Communicating rami: Connect the spinal nerve with the sympathetic trunk.



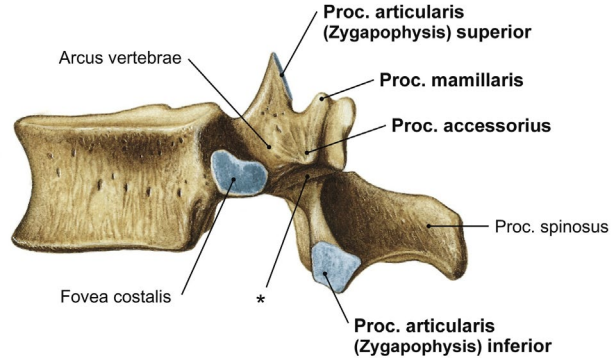


Foramen intervertebrale

Intervertebral Foramen

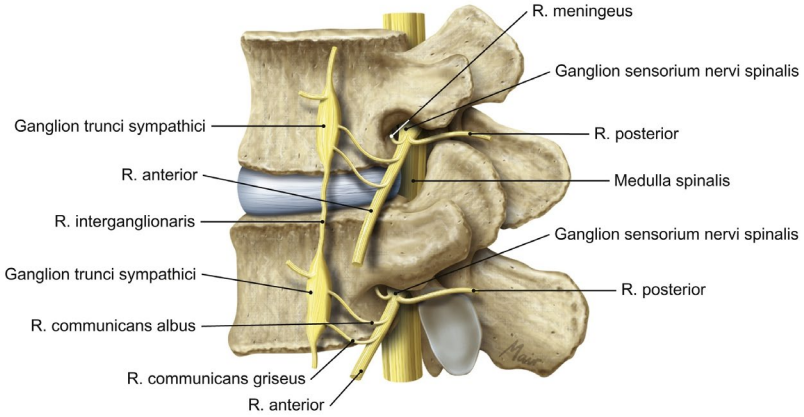
- Fortil og bagtil: Afgrænses af hvirvellegemer/discus og facetleddet.
  Anterior and posterior boundaries: Vertebral bodies/disc and facet joint.

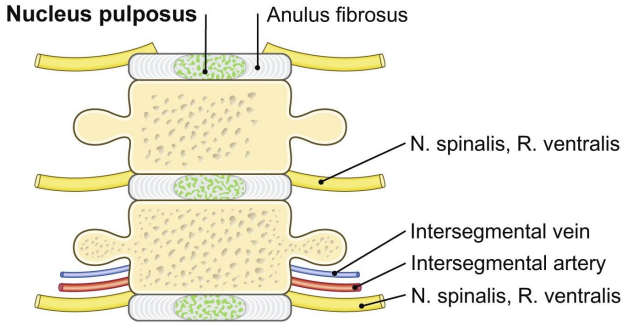
- Opad og nedad: Afgrænses af tilstødende pedikler.
  Superior and inferior boundaries: Adjacent vertebral pedicles.

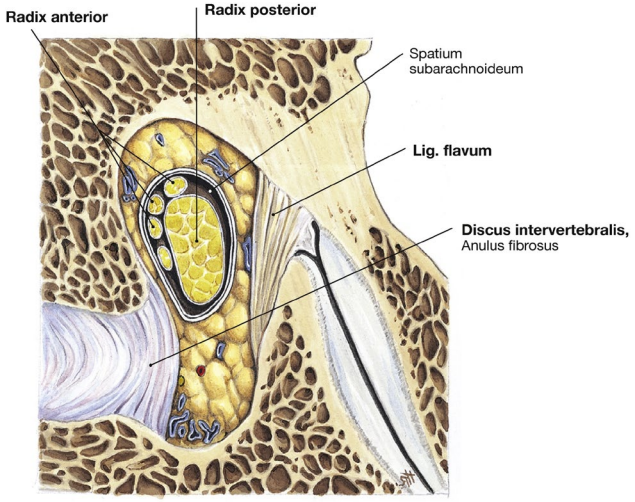


Nervestrukturer: Spinalnerven og spinalgangliet ligger i relation til foramen.
Neural structures: The spinal nerve and dorsal root ganglion are closely related to the foramen.

Små blodkar og bindevæv følger nerverne.
Small vessels and connective tissue accompany the nerves.





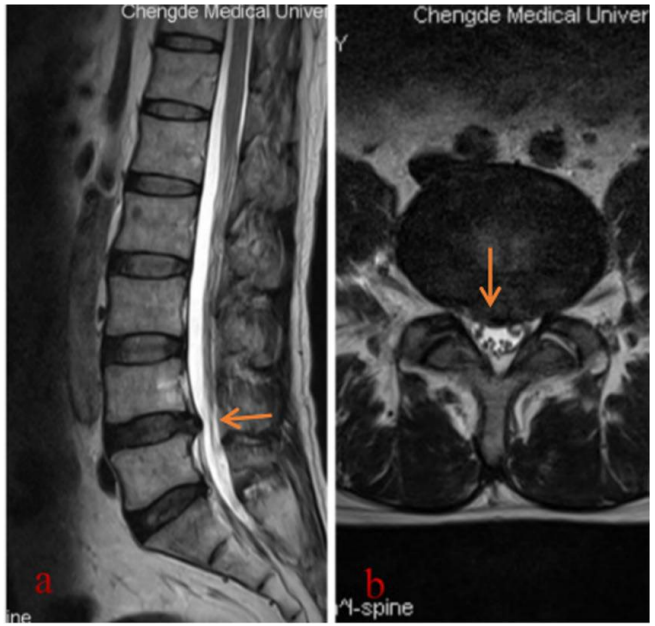


Diskus og nerverødder – Anatomiske relationer

Intervertebral Discs and Spinal Nerve Roots

En patient har en posterolateral diskusprolaps mellem L4 og L5. Hvilken nerverod er mest sandsynligt påvirket?

A patient has a posterolateral disc herniation at L4–L5. Which nerve root is most likely affected?



- Lumbalregionen: Ved L4–L5 passerer L4-roden ud gennem foramen, mens L5-roden fortsætter nedad i spinalkanalen.

Lumbar region: At L4–L5, the L4 root exits through the foramen, while the L5 root descends within the vertebral canal.

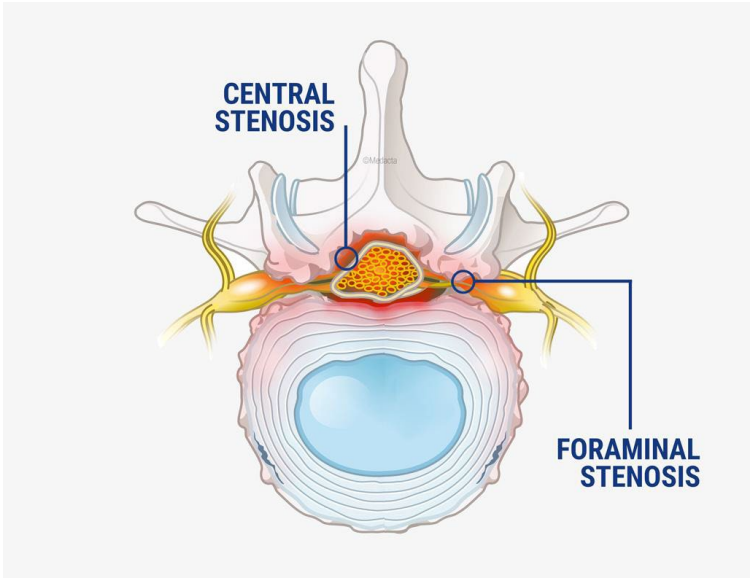
- En posterolateral diskusprolaps ved L4–L5 kan derfor påvirke den passerende L5-nerverod.

A posterolateral L4–L5 disc herniation therefore affect the traversing L5 nerve root.

Spinalstenose – spinalkanal versus foramen intervertebrale

Spinal stenosis — central versus foraminal

- **Central canal stenosis**
  - Forsnævring i spinalkanalen
    Narrowing in the vertebral canal
  - Kan påvirke flere nerverødder
    May affect several nerve roots



- **Foraminal stenosis**
  - Forsnævring ved nerverodens udgang
    Narrowing at the nerve exit
  - Påvirker ofte den udtrædende nerverod
    Often affects the exiting nerve root

Symptomer afhænger af, hvor forsnevringen sidder.

Symptoms depend on where the narrowing is located.

Part II — Cervikalregionens anatomi

Part II – Regional Anatomy of the Cervical Spine

Læringsmål

- Beskrive de vigtigste anatomiske relationer mellem nakkemuskler, cervikale spinalnerver og a. vertebralis.
  Describe the principal anatomical relationships among posterior cervical muscles, spinal nerves, and the vertebral artery.

- Forklare plexus brachialis' relation til halscolumna, scalenermusklerne og de subklaviske kar.
  Explain the relationship of the brachial plexus to the cervical spine, scalene muscles, and subclavian vessels.

- Identificere centrale C5–T1 dermatomer, myotomer og spinalreflekser.
  Identify key C5–T1 dermatomes, myotomes, and spinal reflexes.

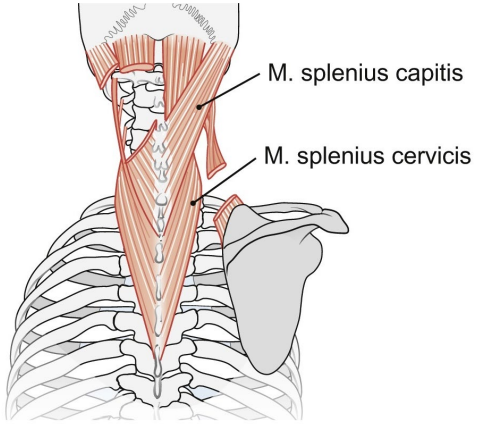
Cervikalregionens anatomiske relationer

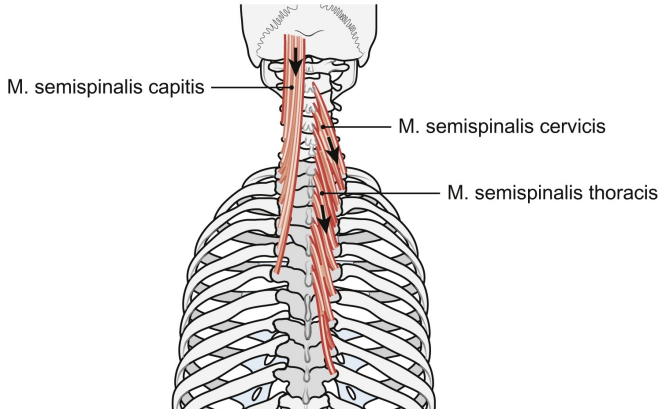
Anatomical Relationships of the Cervical Region

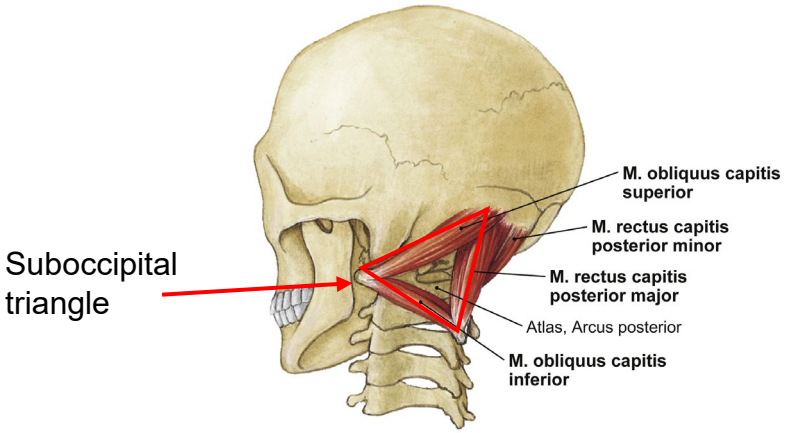
- Dybe nakkemuskler: ***Mm. splenii*** og ***mm. semispinales*** ligger i forskellige muskelplaner bag halshvirvlerne.
Deep neck muscles: Splenius and semispinalis muscles occupy different layers posterior to the cervical vertebrae.

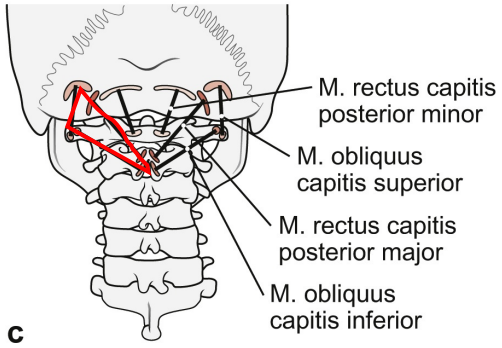
- Suboccipitale muskler: Fire korte muskler ligger dybt mellem os occipitale, atlas (C1) og axis (C2).
Suboccipital muscles: Four short muscles lie deeply between the occipital bone, atlas (C1), and axis (C2).

- Anatomiske relationer: De suboccipitale muskler danner en trekant, der har tæt relation til ***a. vertebralis*** og ***n. suboccipitalis*** (C1).
Anatomical relationships: The suboccipital muscles form a triangle closely associated with the vertebral artery and suboccipital nerve (C1).









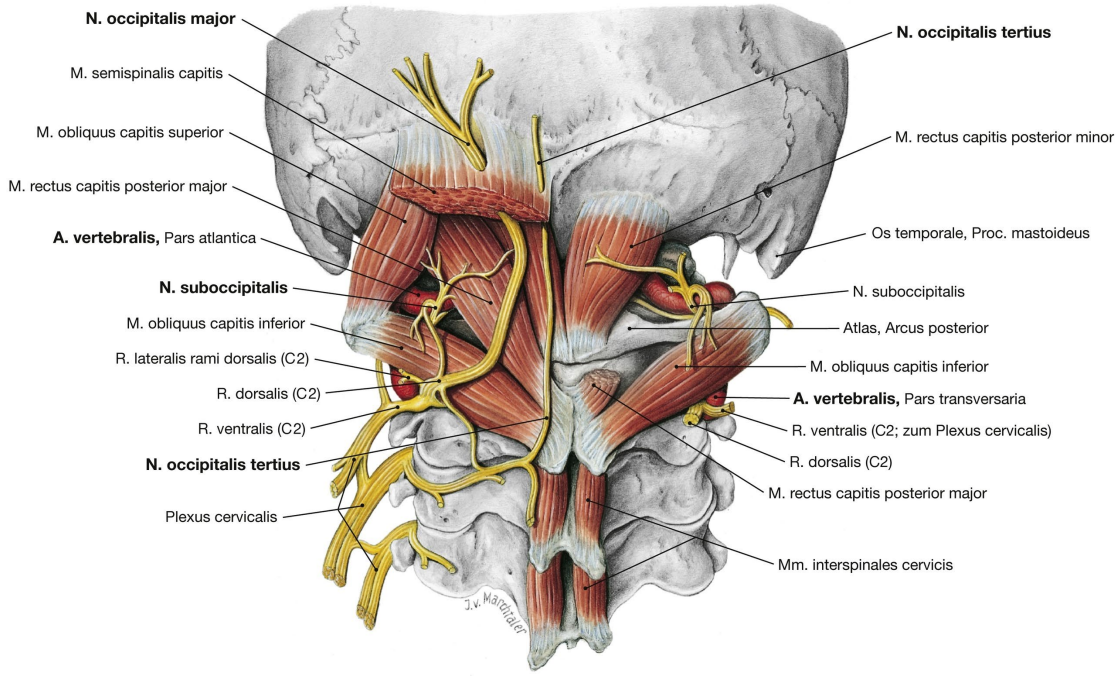
Suboccipitalregionens nerve- og karrelationer

Neurovascular Relationships of the Suboccipital Region

- Hvirvelarterien (***a. vertebralis***): Løber i en fure på atlas' bageste bue gennem den suboccipitale trekant.
Vertebral artery: Runs in a groove on the posterior arch of the atlas through the suboccipital triangle.

- Suboccipitalnerven (***n. suboccipitalis, C1***): Den bageste gren af C1, som primært innerverer de korte suboccipitale muskler.
Suboccipital nerve: The posterior ramus of C1, primarily supplying the short suboccipital muscles.

- Store occipitalnerve (***n. occipitalis major, C2***): Løber under m. obliquus capitis inferior og fortsætter gennem nakkens muskler mod baghovedets hud.
Greater occipital nerve: Passes inferior to obliquus capitis inferior and continues through posterior neck muscles toward the occipital scalp.



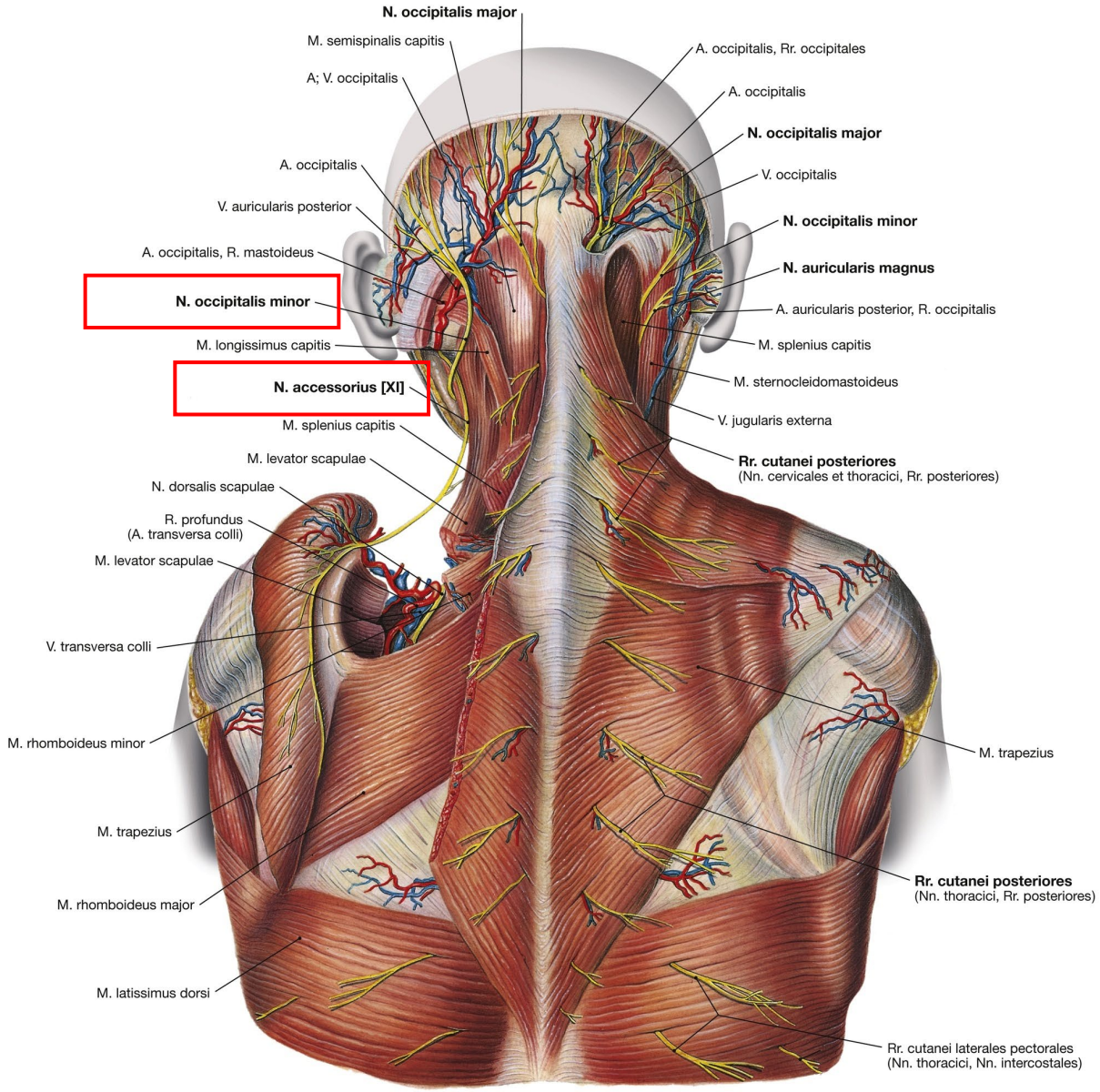
Nakkeregionens kutane nerver

Cutaneous Nerves of the Posterior Cervical Region

- Segmental hudinnervation: De bageste spinalnervegrene (rr. posteriores) afgiver kutane grene til nakkens og den øvre rygs hud.
Segmental cutaneous innervation: Posterior spinal nerve rami supply the skin of the posterior neck and upper back.

- Lille occipitalnerve (***n. occipitalis minor***, C2–C3): Udspringer fra plexus cervicalis og forsyner huden bag øret og på den laterale del af baghovedet.
Lesser occipital nerve: Arises from the cervical plexus and supplies the skin behind the ear and lateral occipital scalp.

- Accessorisk nerve (***n. accessorius***, XI): Løber gennem nakkens laterale region og innerverer m. trapezius.
Accessory nerve: goes through the lateral neck and supplies the trapezius muscle.

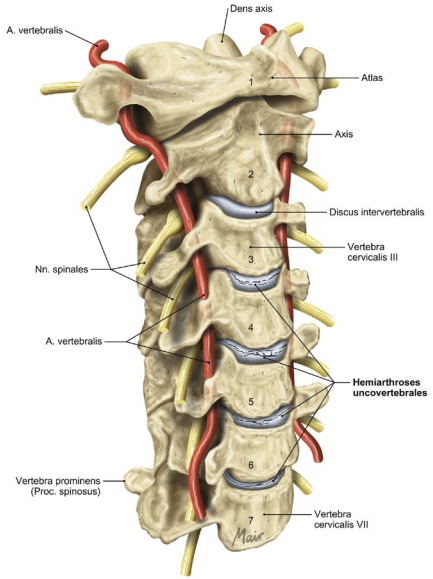


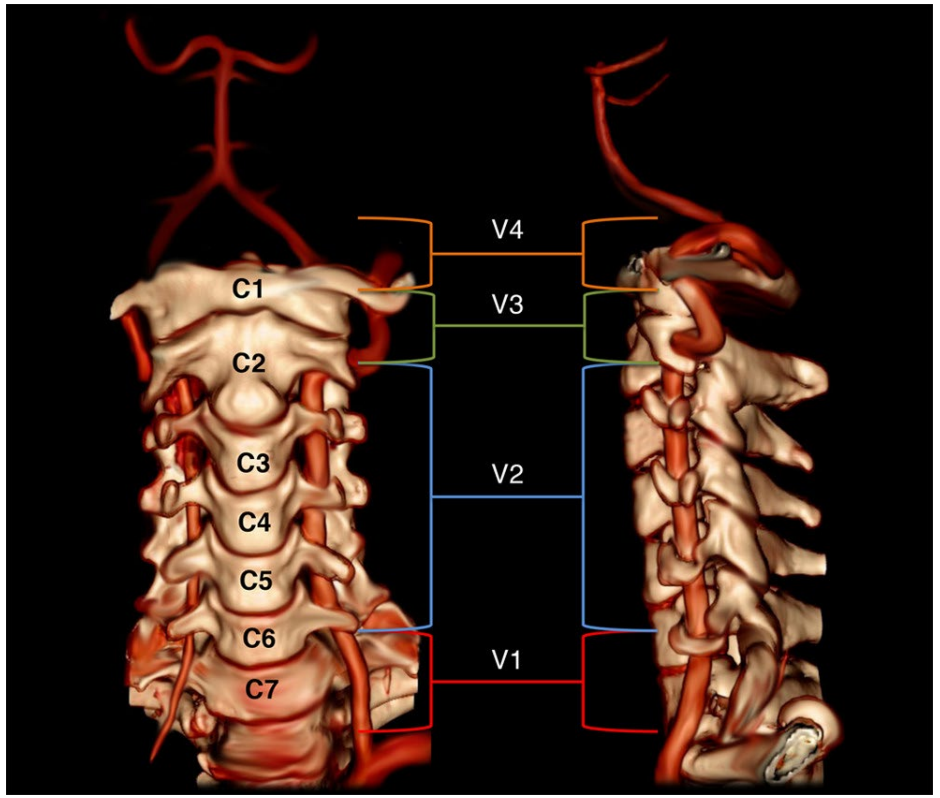
Hvirvelarteriens relation til halscolumna

Vertebral Artery and Cervical Spine

- Forløb: A. vertebralis passerer sædvanligvis gennem tværtapshullerne C6–C1.
Course: The vertebral artery usually passes through the transverse foramina of C6–C1.

- Øvre halsregion: Arterien passerer over atlas' bageste bue før indtræden i kraniet.
Upper cervical region: The artery passes over the posterior arch of atlas before entering the cranium.





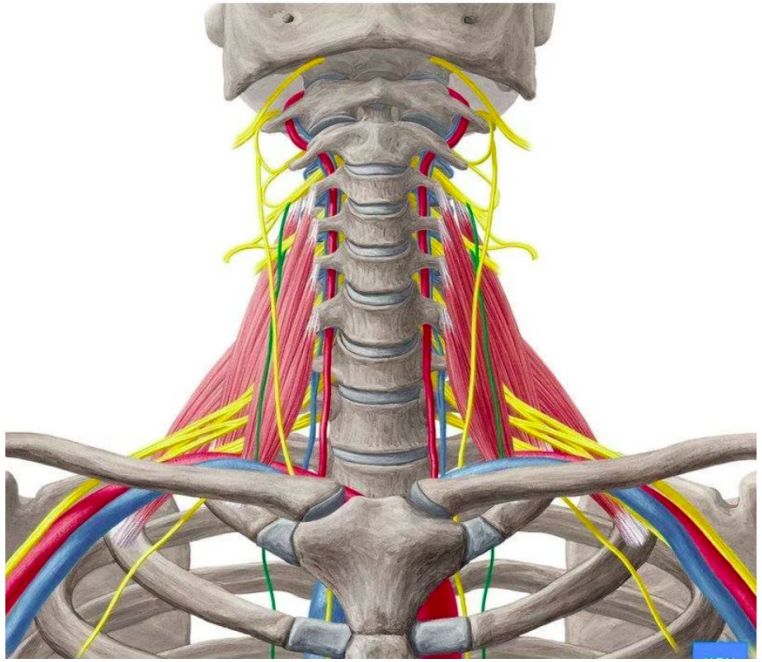
Plexus brachialis og anatomiske relationer

Brachial Plexus and Anatomical Relationships

- Oprindelse: Plexus brachialis dannes primært af de forreste spinalnervegrene fra C5–T1.
Origin: The brachial plexus is formed primarily by the anterior rami of C5–T1.

- Forløb: Nerverne passerer mellem m. scalenus anterior og m. scalenus medius mod overekstremiteten.
Course: The nerves pass between the anterior and middle scalene muscles toward the upper limb.

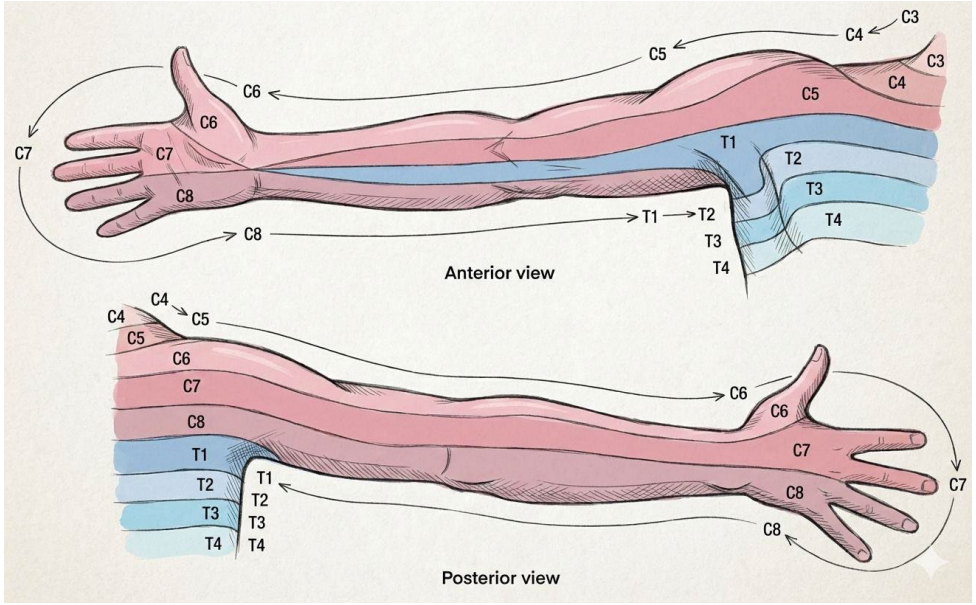
- Karrelation: A. subclavia passerer sammen med plexus mellem scalenermusklerne, mens v. subclavia løber foran m. scalenus anterior.
Vascular relationship: The subclavian artery passes between the scalene muscles with the plexus, while the subclavian vein passes anterior to anterior scalene.



Cervikale nerverødder: Motorisk og sensorisk fordeling

Cervical Nerve Roots – Motor and Sensory Distribution

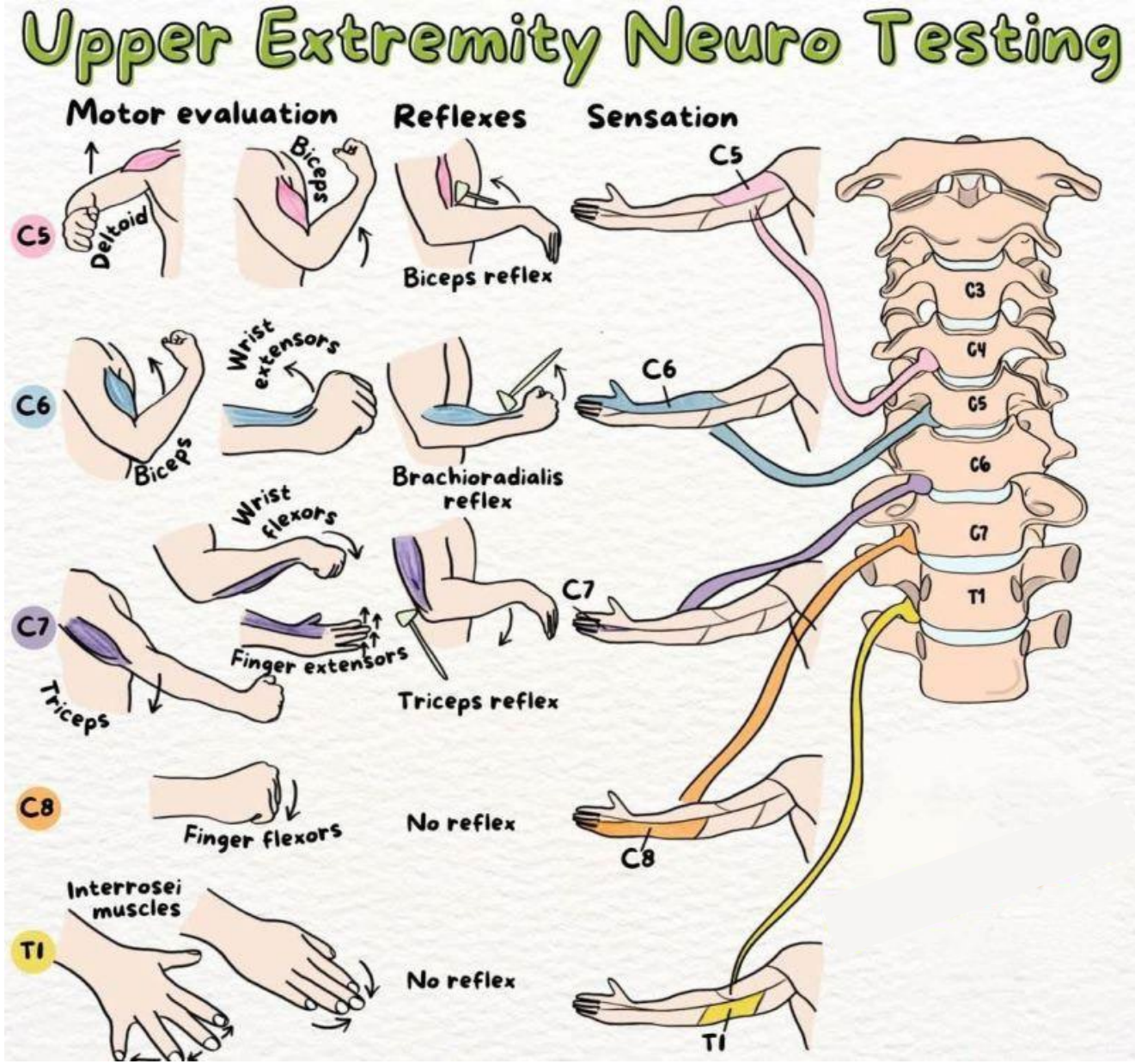
Nerverod	Motorisk funktion	Sensorisk område
C5	Skulderabduktion *Shoulder abduction*	Laterale overarm *Lateral upper arm*
C6	Albuefleksion, håndledsekstension *Elbow flexion, wrist extension*	Tommel *Thumb*
C7	Albueekstension *Elbow extension*	Langfinger *Middle finger*
C8	Fingerfleksion *Finger flexion*	Lillefinger *Little finger*
T1	Fingerspredning *Finger abduction*	Mediale underarm *Medial forearm*



Posterior and anterior views of the upper limb dermatomes.



Bicepsrefleks Biceps reflex	C5–C6
Brachioradialisrefleks Brachioradialis reflex	C5–C6, især C6
Tricepsrefleks Triceps reflex	C7–C8, især C7


Upper Extremity Neuro Testing
Motor evaluation
Reflexes
Sensation
C5
Deltoid
Biceps
Biceps reflex
C6
Wrist extensors
Biceps
Brachioradialis reflex
C7
Wrist flexors
Finger extensors
Triceps
Triceps reflex
C8
Finger flexors
No reflex
T1
Interrosei muscles
No reflex
C3
C4
C5
C6
C7
T1

Part III — Thorakalregionens anatomi

Part III – Regional Anatomy of the Thoracic Spine

Læringsmål

- Beskrive de thorakale spinalnervers og den sympatiske grænsestrengs relation til brysthvirvelsøjlen.
  Describe the relationships of thoracic spinal nerves and the sympathetic trunk to the thoracic spine.

- Forklare interkostalnervernes og blodkarrenes anatomiske forløb i relation til ribbenene.
  Explain the anatomical course of intercostal nerves and vessels in relation to the ribs.

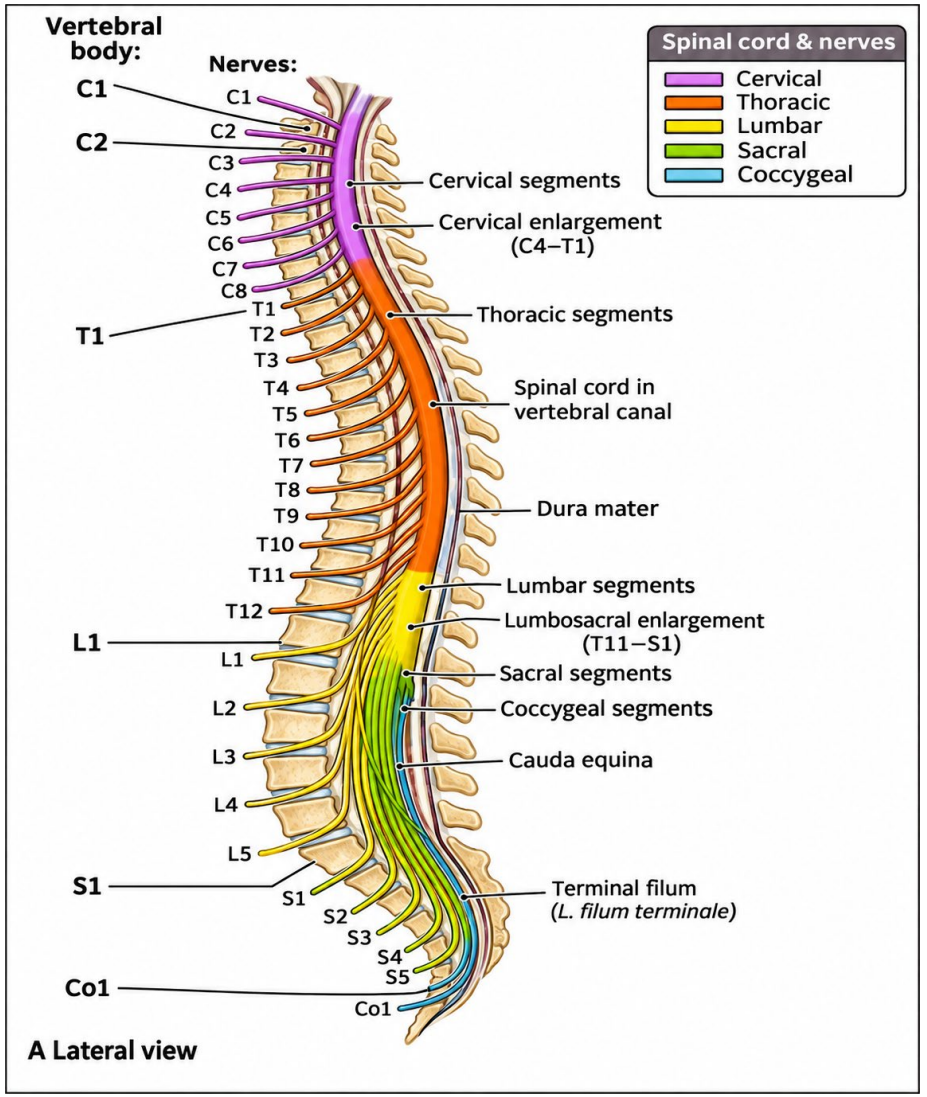
- Identificere thorakale dermatomer og anvende deres segmentale fordeling til at forstå kliniske symptomer.
  Identify thoracic dermatomes and apply their segmental distribution to understand clinical symptoms.

Spinalnervernes udgangsniveauer

Spinal Nerve Exit Levels

- Cervikalt: C1–C7 udgår over den tilsvarende hvirvel; C8 mellem C7 og T1.
  Cervical: C1–C7 exit above the corresponding vertebrae; C8 exits between C7 and T1.

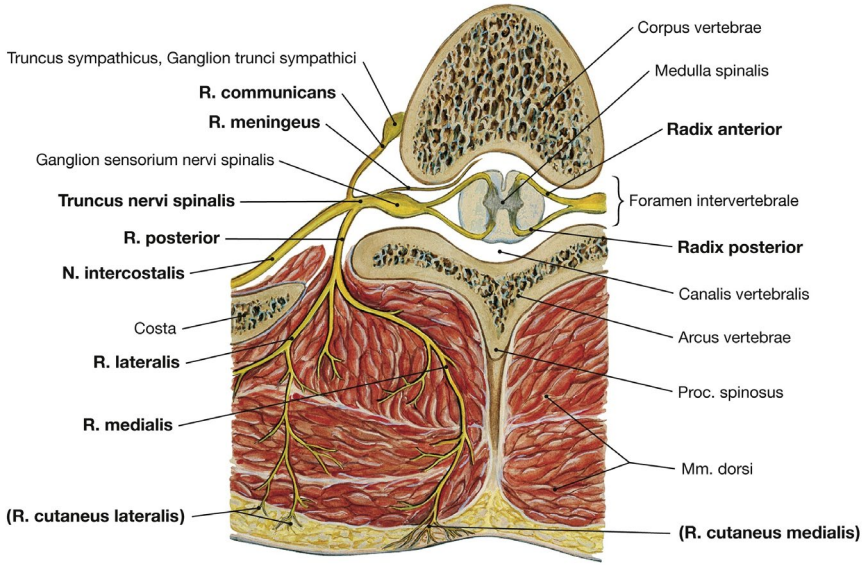
- Fra T1 og kaudalt: Nerverne udgår under den tilsvarende hvirvel.
  From T1 caudally: Spinal nerves exit below the corresponding vertebrae.

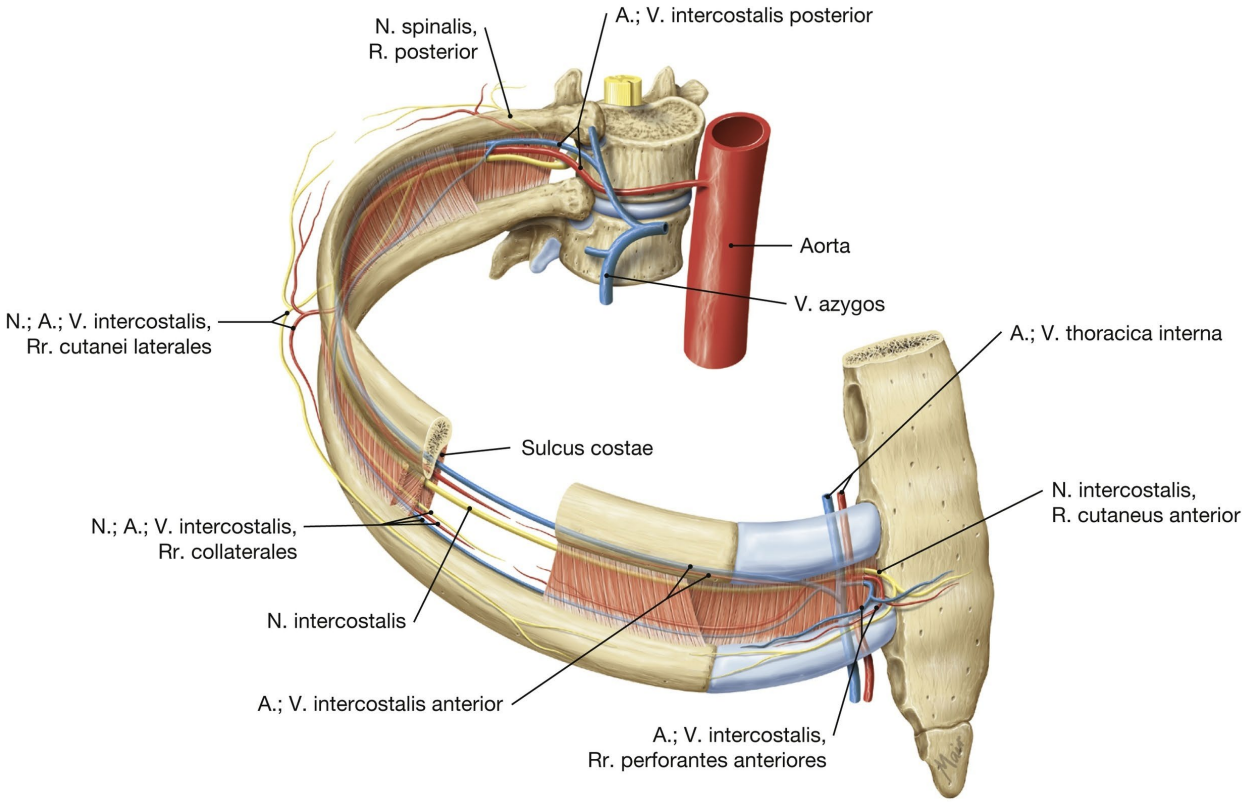


Spinalnervernes relationer
Spinal Nerve Relationships

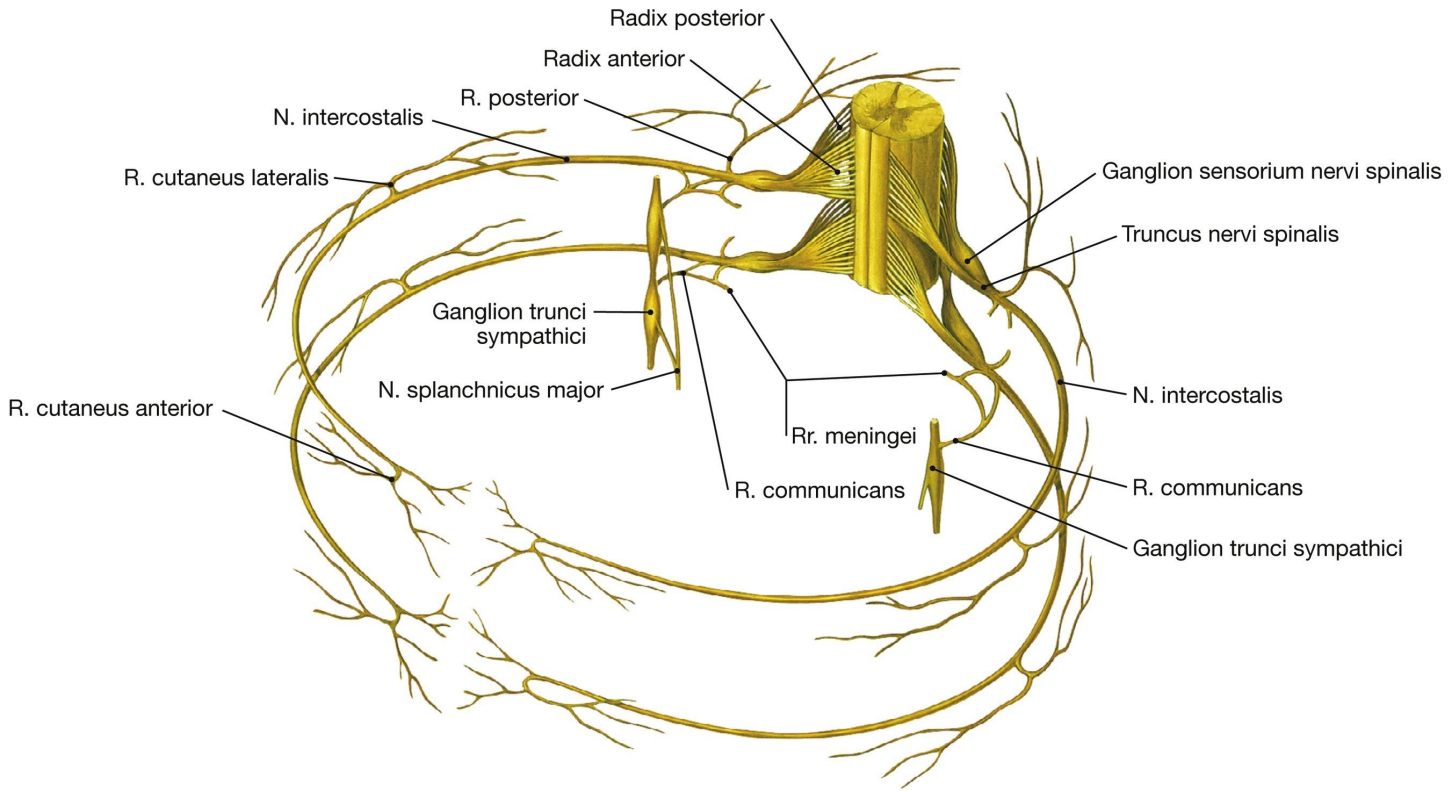
- Bageste grene: Løber mod de paravertebrale strukturer og ryggens hud.
Posterior rami: Travel toward paravertebral structures and the skin of the back.

- Forreste grene: Fortsætter overvejende som interkostalnerver i thorakalregionen.
Anterior rami: Continue predominantly as intercostal nerves in the thoracic region.





Thorakalregionens spinalnerver og sympatiske grænsestreng
Thoracic Spinal Nerves and Sympathetic Trunk



- Sympatisk grænsestreng: Truncus sympathicus ligger på begge sider af brysthvirvelsøjlen tæt ved ribbenshovederne.
Sympathetic trunk: The sympathetic trunk lies on either side of the thoracic spine, close to the heads of the ribs.

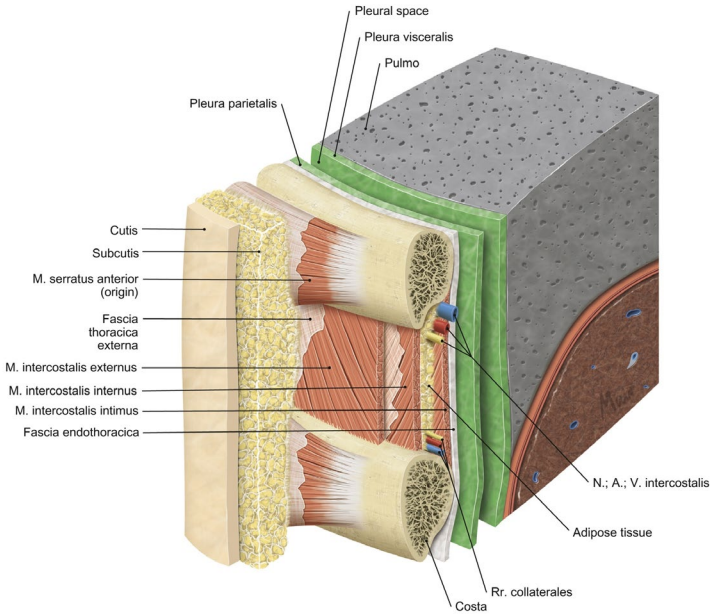
- Forbindelser: ***Rr. communicantes*** forbinder spinalnerverne med den sympatiske grænsestreng.
Connections: Communicating rami connect spinal nerves with the sympathetic trunk.

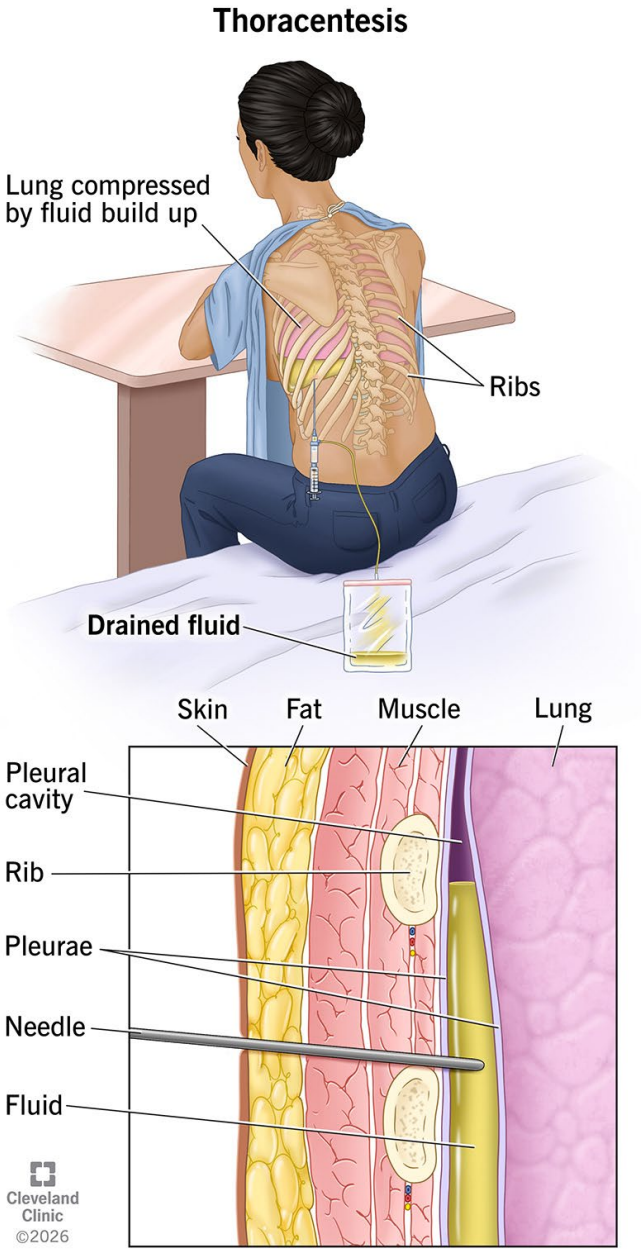
Interkostalnervernes og karrenes relation til ribbenene
Intercostal Neurovascular Relationships

- Forløb: Nn. intercostales følger ribbenene sammen med de interkostale blodkar.
Course: Intercostal nerves travel alongside the intercostal blood vessels.

- Placering: Kar og nerver ligger i sulcus costae på ribbenets nedre kant, mellem mm. intercostales interni og intimi.
Location: The neurovascular bundle lies in the costal groove, between the internal and innermost intercostal muscles.

- Rækkefølge: Fra kranialt til kaudalt: vene, arterie og nerve (V–A–N).
Order: From superior to inferior: vein, artery and nerve (V–A–N).





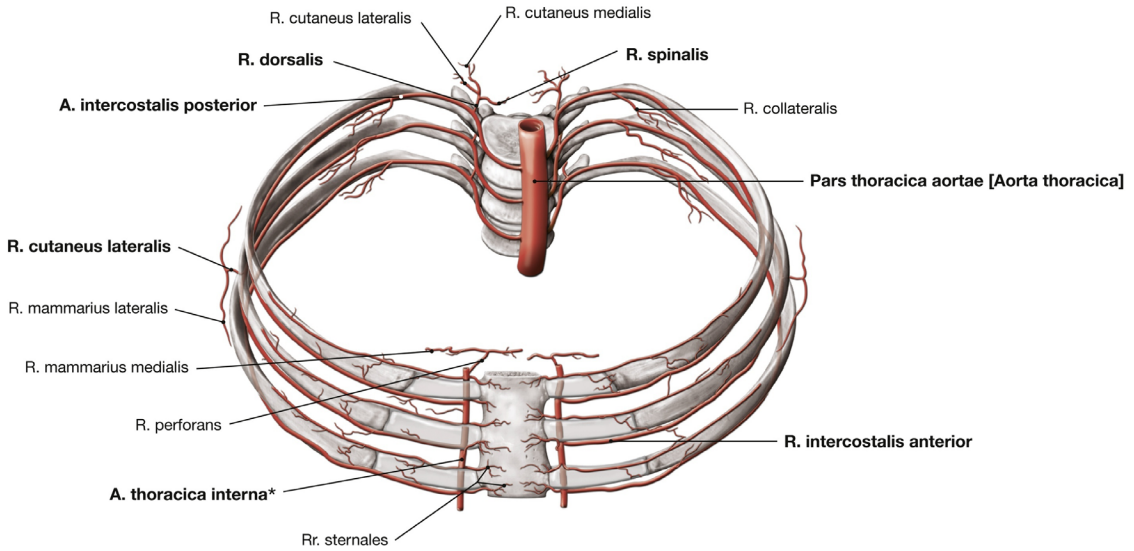
Thorakalregionens segmentale karforsyning

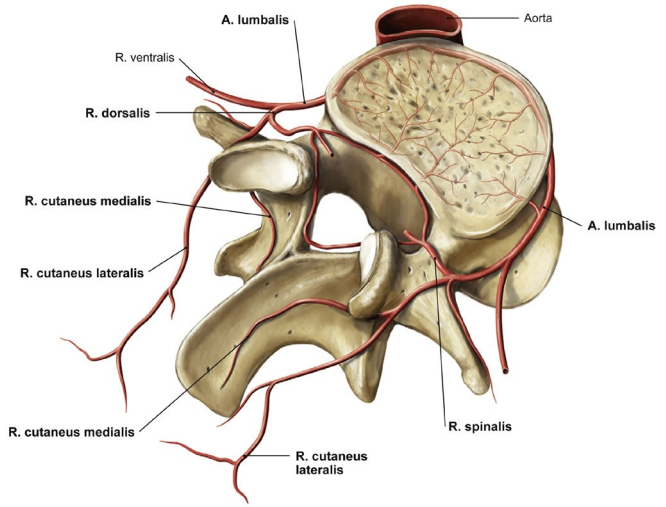
Segmental Vascular Supply of the Thoracic Vertebral Region

- Segmentale arterier: Aa. intercostales posteriores forløber fra den bageste thoraxvæg mod ribbenene og afgiver grene nær rygsøjlen.
Segmental arteries: Posterior intercostal arteries supply the thoracic wall and give off branches near the vertebral column.

- Dorsale grene: Rr. dorsales forsyner bl.a. de dybe rygmuskler og tilstødende knoglestrukturer.
Dorsal branches: Supply the deep back muscles and adjacent skeletal structures.

- Spinale grene: Rr. spinales passerer gennem foramina intervertebralia og bidrager til karforsyningen i vertebralkanalen.
Spinal branches: Enter through the intervertebral foramina and contribute to vascular supply within the vertebral canal.





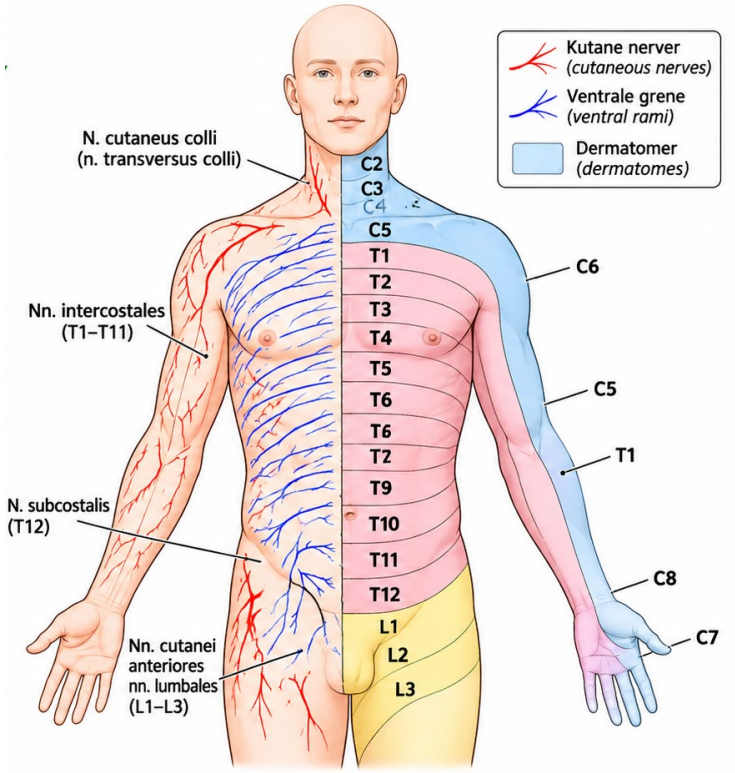
Thorakale dermatomer

Thoracic Dermatomes

- Segmental innervation: De thorakale spinalnerver (T2–T12) forsyner huden på brystkassen og bugvæggen i omtrent bælteformede områder.
Segmental innervation: Thoracic spinal nerves (T2–T12) supply the skin of the chest and abdominal wall in approximately band-shaped regions.

- Vigtige anatomiske landemærker:
  - T4 – brystvorte; *T4 – nipple*
  - T6 – sværdformet tap; *T6 – xiphoid process*
  - T10 – navle. *T10 – umbilicus.*

- Nervernes forløb: De forreste grene fortsætter som nn. intercostales (T1–T11) og n. subcostalis (T12), der afgiver kutane grene til kropsvæggen.
Nerve pathways: The anterior rami continue as intercostal nerves (T1–T11) and the subcostal nerve (T12), giving rise to cutaneous branches of the body wall.

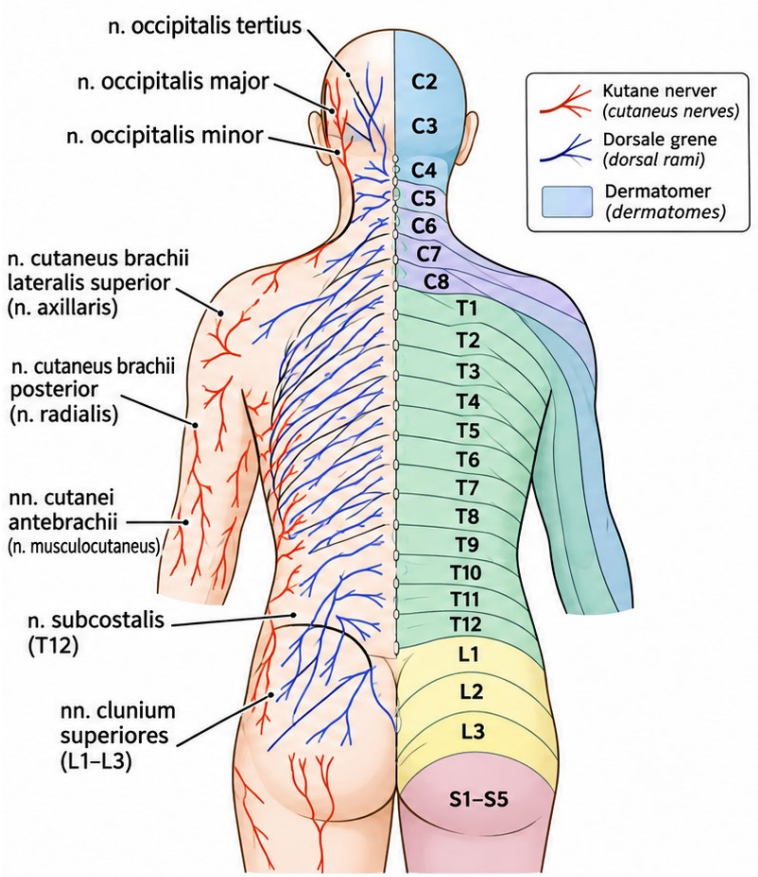


Thorakale dermatomer

Thoracic Dermatomes

- Dorsale grene: Rr. posteriores innerverer de dybe rygmuskler og afgiver kutane grene til huden på ryggen.
Dorsal rami: Supply deep back muscles and give rise to cutaneous branches supplying the skin of the back.

- Regional fordeling: De mediale og laterale kutane grene forsyner forskellige områder af ryggen; i den øvre thorakalregion dominerer de mediale grene, mens de laterale grene bliver vigtigere længere kaudalt.
Regional distribution: Medial cutaneous branches predominate in the upper thoracic back, while lateral branches become more prominent at lower thoracic levels.



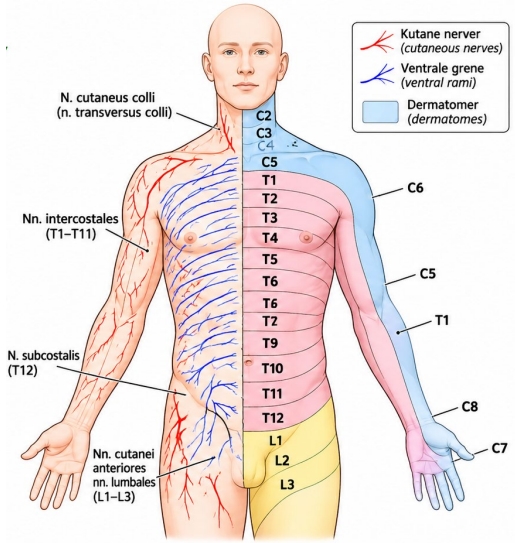
Klinisk eksempel: Herpes zoster (helvedesild)

Clinical Example: Herpes Zoster (Shingles)

Hvorfor er udslættet typisk lokaliseret til et bælteformet område på én side af brystkassen?

Why is the rash typically confined to a band-shaped region on one side of the thorax?





Reaktivering af varicella-zoster-virus i et sensorisk ganglion (ganglion spinale) kan forårsage udslæt i det tilhørende dermatom, typisk på én side af kroppen.

Reactivation of varicella-zoster virus in a sensory ganglion (dorsal root ganglion) can produce a rash in the associated dermatome, typically on one side of the body.

Part IV — Lumbal- og sakralregionens anatomi

Part IV – Regional Anatomy of the Lumbar and Sacral Spine

Læringsmål

- Beskrive de vigtigste relationer mellem lændehvirvlerne, rygmusklerne, m. psoas major og m. quadratus lumborum.
  Describe the principal relationships among lumbar vertebrae, back muscles, psoas major, and quadratus lumborum.

- Forklare plexus lumbalis og plexus sacralis' anatomiske placering og deres relation til de omgivende muskler.
  Explain the anatomical positions of the lumbar and sacral plexuses and their relationships to surrounding muscles.

- Identificere centrale L2–S1 dermatomer, myotomer og reflekser samt relatere dem til spinalnerverødderne.
  Identify key L2–S1 dermatomes, myotomes, and reflexes and relate them to spinal nerve roots.

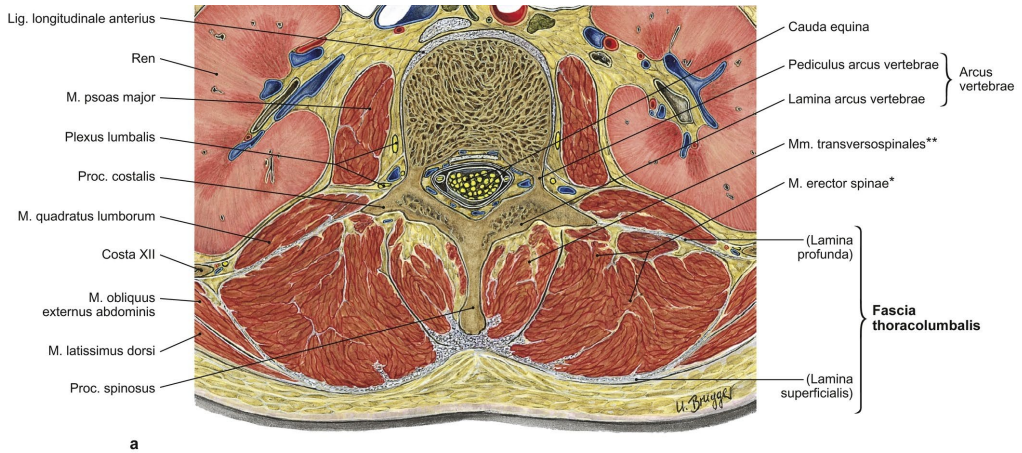
Lumbalregionens muskulære relationer

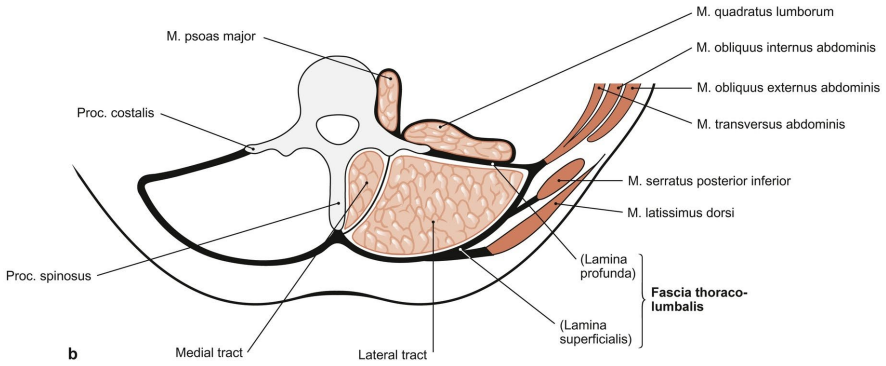
Muscular Relationships of the Lumbar Region

- Bagtil: M. erector spinae og m. multifidus ligger posteriort for lændehvirvlerne og bidrager til ekstension og stabilisering, og omsluttes af fascia thoracolumbalis.
Posteriorly: Erector spinae and multifidus lie behind the lumbar vertebrae and contribute to extension and stabilization, enclosed by the thoracolumbar fascia.

- Lateralt: M. quadratus lumborum forbinder bækkenet, lændehvirvlerne og 12. ribben.
Laterally: Quadratus lumborum connects the pelvis, lumbar vertebrae, and 12th rib.

- Fortil: M. psoas major ligger anterolateralt for lændehvirvlerne og indeholder plexus lumbalis.
Anteriorly: Psoas major lies anterolateral to the lumbar vertebrae and contains the lumbar plexus.



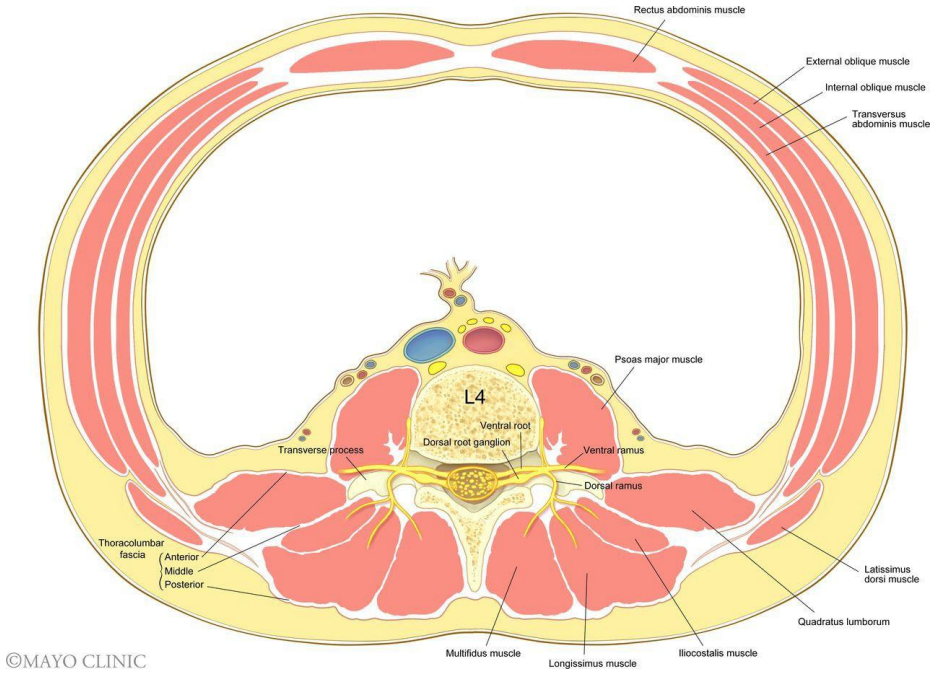


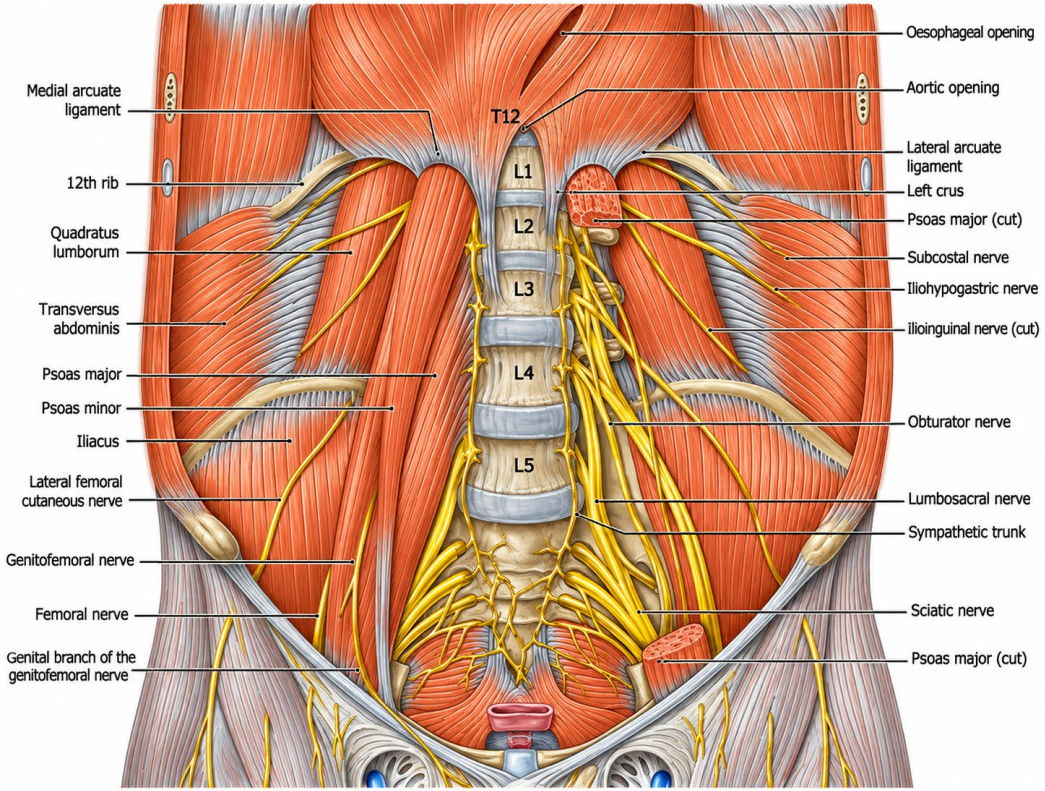
Lumbalregionens muskulære relationer

Muscular Relationships of the Lumbar Region

- Plexus lumbalis dannes i den bageste del af m. psoas major.
The lumbar plexus forms within the posterior part of psoas major.

- Aorta abdominalis og v. cava inferior ligger foran lændehvirvelsøjlen og medialt for m. psoas major.
The abdominal aorta and inferior vena cava lie anterior to the lumbar spine and medial to psoas major.





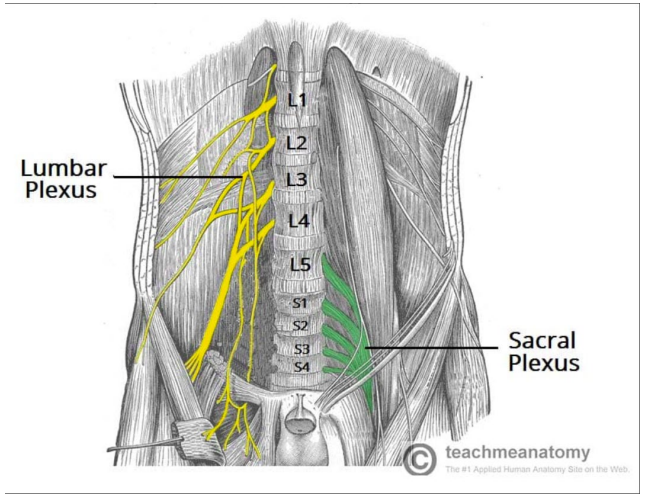
Plexus lumbalis og relationen til m. psoas major

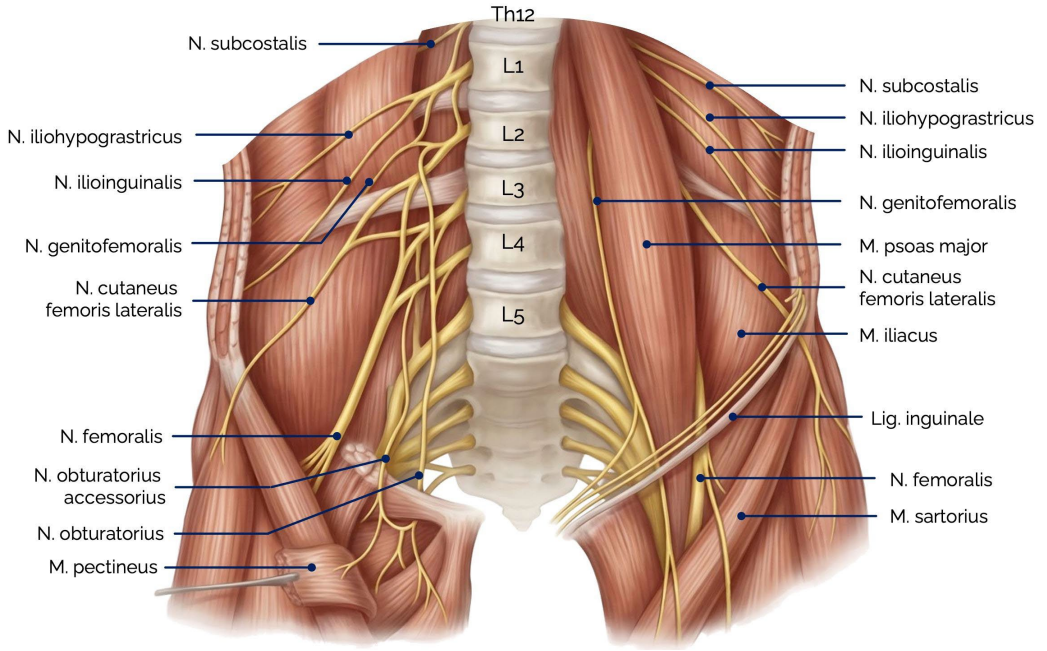
Lumbar Plexus and Its Relationship to Psoas Major

- Oprindelse: Plexus lumbalis dannes af de forreste spinalnervegrene fra L1–L4.
Origin: The lumbar plexus is formed by the anterior rami of L1–L4.

- Laterale nervegrene: ***N. femoralis*** (L2–L4) forlader m. psoas major lateralt og fortsætter mod lårets forside.
Lateral branches: The femoral nerve emerges from the lateral border of psoas major and travels toward the anterior thigh.

- Mediale nervegrene***: N. obturatorius*** (L2–L4) forlader musklen medialt og fortsætter gennem bækkenet mod lårets mediale region.
Medial branches: The obturator nerve emerges medially and passes through the pelvis toward the medial thigh.





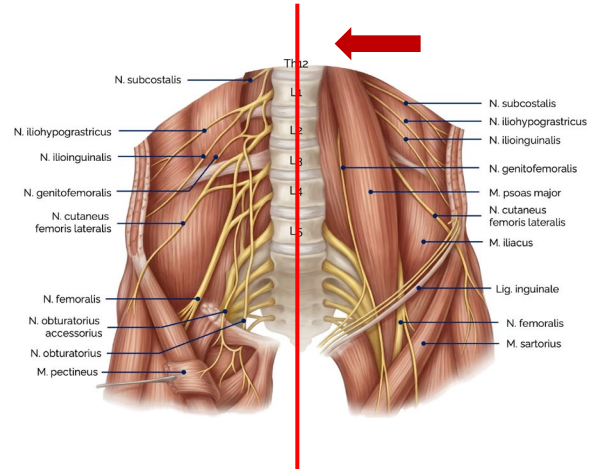
Plexus sacralis og bækkenets anatomiske relationer

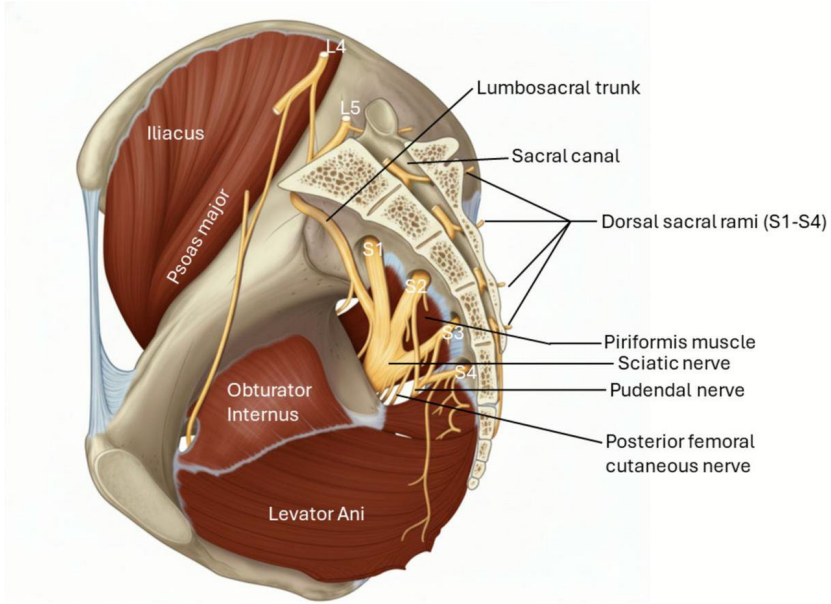
Sacral Plexus and Anatomical Relationships of the Pelvis

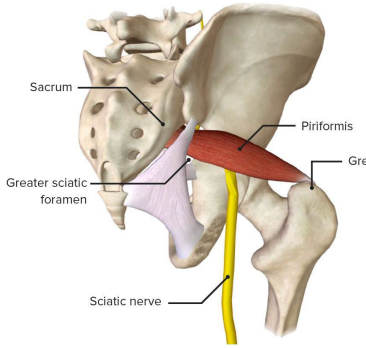
- Plexus sacralis dannes af forreste spinalnervegrene fra L4–S4 og ligger foran m. piriformis.
The sacral plexus is formed by anterior rami of L4–S4 and lies anterior to piriformis.

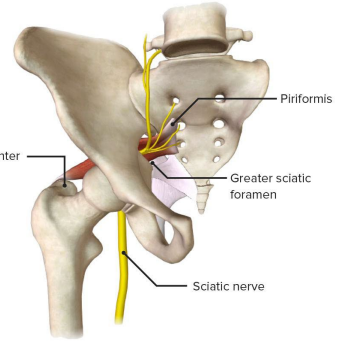
- Nerveforløb: ***N. ischiadicus*** (L4–S3) forlader bækkenet gennem foramen ischiadicum majus, sædvanligvis under m. piriformis.
Nerve course: The sciatic nerve exits the pelvis through the greater sciatic foramen, usually inferior to piriformis.

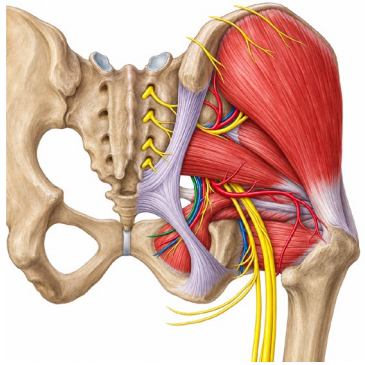
- Relation til underekstremiteten: Plexus sacralis afgiver nerver til sæderegionen, lårets bagside og størstedelen af underbenet og foden.
Lower-limb relationship: The sacral plexus supplies nerves to the gluteal region, posterior thigh, and most of the leg and foot.







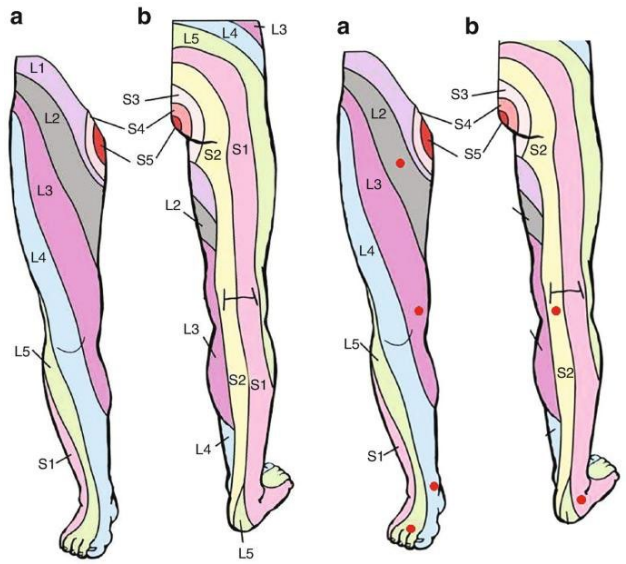


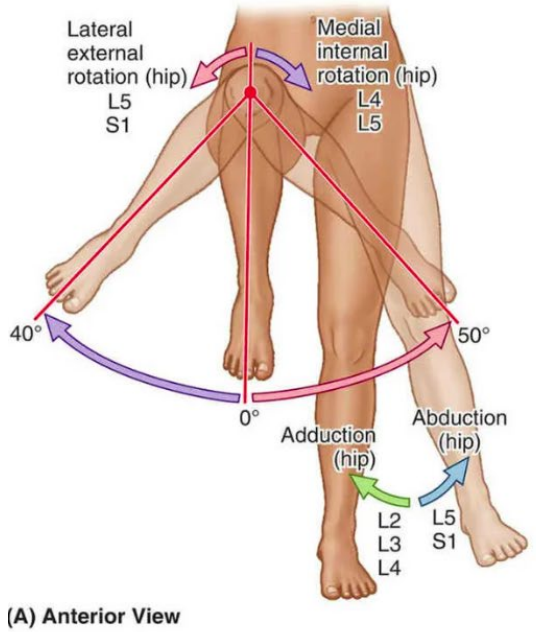


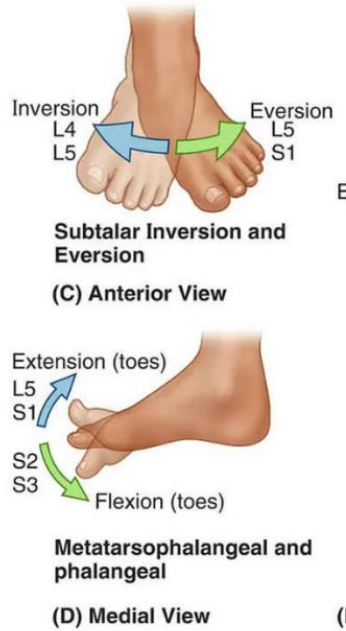
Lumbosakrale nerverødder: Motorik, sensibilitet og reflekser

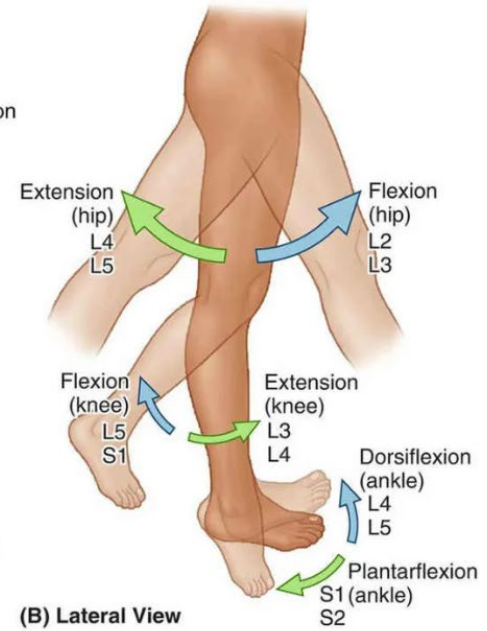
Lumbosacral Nerve Roots: Motor, Sensory and Reflex Functions

Nerverod	Motorisk funktion *Motor function*	Sensorisk område *Sensory region*
L2	Hoftefleksion *Hip flexion*	Forreste lår *Anterior thigh*
L3	Knæekstension *Knee extension*	Mediale knæ *Medial knee*
L4	Dorsalfleksion *Ankle dorsiflexion*	Mediale ankel *Medial ankle*
L5	Storetåekstension *Great-toe extension*	Fodryg, storetå *Dorsum of foot, great toe*
S1	Plantarfleksion *Plantar flexion*	Laterale fod *Lateral foot*

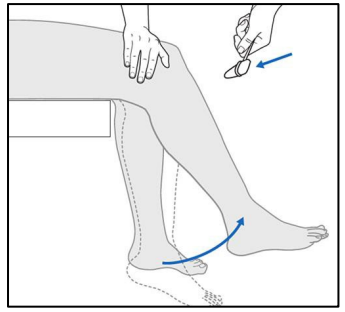


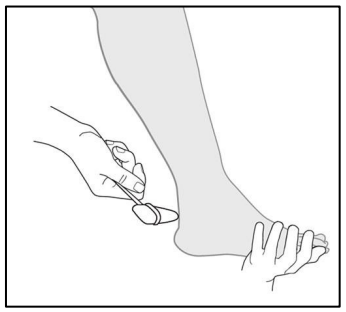






Refleks	Spinalsegmenter	Nerve
Patellarrefleks *Patellar reflex*	L3–L4, især L4	N. femoralis
Achillesrefleks *Achilles reflex*	S1–S2, især S1	N. tibialis





Do not edit
How to change the design



En patient kan ikke ekstendere storetåen normalt. Hvilken nerverod er særligt relevant?
A patient has difficulty extending the great toe. Which nerve root is particularly relevant?

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